

Reciprocal-Tail Rigidity: Theorems, Proofs and Questions

Erdős Problem #243

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ABSTRACT

If strictly increasing positive integers satisfy $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$, their reciprocal sum is irrational. The proof first extracts an exact cubic integer-tail profile, then excludes it by denominator transport, a cubic-field calculation and a finite congruence obstruction. We also study rational reciprocal sums with $a_{n+1}/a_n^2 \rightarrow 1$. An eventual lower bound on the centred integer error forces a Sylvester tail; an LCM record criterion describes the remaining finiteness obligation. The rational-tail analysis starts from Duverney's and Koizumi's frameworks; the bounded-negative theorem extends the eventual-nonnegative descent. The bounded-defect and finite-mass implications, including specified canonical transfers, are formalised in the checked public source. The cubic theorem is presented as an ordinary proof with formalised components, not as an assembled Lean theorem. We give ordinary proofs of the prime-power supply needed by the record-energy argument and of an exact maximal-gap coefficient for a static modulus model. The distinction between that model and the exact tail dynamics leaves the unrestricted question unresolved.

1 Introduction

Throughout, $z_+ = \max(z, 0)$. Statements about a tail permit deletion of a finite prefix, but the index in a specified higher-order rate is retained. In particular, $1 + 3/n + o(n^{-3})$ must not be silently replaced by the same formula in a shifted index. Positive integer hypotheses refer to ordinary integers, not merely positive real states.

Sylvester's sequence $2, 3, 7, 43, 1807, \dots$ is generated by $a_n = a_{n-1}^2 - a_{n-1} + 1$ and has reciprocal sum 1; the recurrence is equivalent to the telescoping identity

$$\frac{1}{a_{n-1} - 1} = \frac{1}{a_{n-1}} + \frac{1}{a_n - 1}.$$

Consequently the tail beginning with $1/a_{n-1}$ equals $1/(a_{n-1} - 1)$. Erdős Problem #243 asserts that this is the only way a sequence of that growth can have a rational reciprocal sum:

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Problem 1.1 (Erdős #243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

See [5, p. 64] and [6, p. 105]. The supplied catalogue snapshot labels it open [13]; no fresh catalogue-status verification is claimed for this revision. Below we index the same recurrence as $a_{n+1} = a_n^2 - a_n + 1$, which is the shift the formal sources use; the two forms are the same statement. Write $\text{syl}(a) = a^2 - a + 1$, the [Sylvester successor](#).

The hypothesis is asymptotic and the conclusion is exact. Clearing a rational tail as C_n/D_n exposes that distinction in the integer $E_n = D_n - (a_n - 1)C_n$: a Sylvester tail has $E_n = 0$, while quadratic growth only gives $E_n/C_n \rightarrow 0$. Integrality alone cannot turn this relative estimate into $E_n = 0$ when C_n grows. The exact dynamics retain extra information: old divisors persist, and they constrain which numerator heights the orbit can visit. This is the arithmetic information used by the bounded-negative and record-crossing arguments.

Section 2 carries out that conversion at one prescribed rate and settles the assertion of Problem 1.1 on the resulting class of sequences. Section 3 places the theorem against the published sufficient conditions, states the evidence class of every result here, and guides the rest of the note.

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2 Irrationality at the cubic rate

The short note treats the cubic-rate theorem as a further consequence. This section supplies its full polynomial and number-field argument. The canonical integer state is constructed first; the general error coordinates are developed in Section 4.

Theorem 2.1 (cubic-rate irrationality). *If strictly increasing positive integers satisfy $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$, then $\sum_n 1/a_n$ is irrational.*

Theorem 2.2 (rising-factorial cubic exclusion). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{Z}_{>0}$ satisfy*

$$C_{n+1} = a_n C_n - D_n, \quad D_{n+1} = a_n D_n. \tag{1}$$

Then for every $A \in \mathbb{Q}_{>0}$ and every $B \in \mathbb{Q}$,

$$\liminf_{X \rightarrow \infty} \frac{\#\{n \leq X : C_n \neq A n(n+1)(n+2) + B\}}{X} > 0.$$

In particular no such orbit satisfies $C_n = A n(n+1)(n+2) + B$ for all large n .

The irrationality theorem needs only exclusion of eventual equality. Theorem 2.2 proves more: the exceptional set has strictly positive lower density. Equivalently, no subsequence of prefix lengths can have an exceptional proportion tending to zero. The

density constant may depend on the orbit and on the proposed cubic. The two restrictions that meet on the same cubic are integrality, which supplies the rising-factorial shape and the scale, and the denominator transport, which imposes an arithmetic square condition; the scale calculation and the two residue words that close the argument can be checked on the page.

The canonical integer tail. Suppose $\sum_{n \geq 1} 1/a_n = p/q$ with $p, q \geq 1$, and write $x_n = \sum_{k \geq n} 1/a_k$ for the tail. Put

$$D_n = q \prod_{k < n} a_k, \quad C_n = D_n x_n.$$

Then D_n is a positive integer, and

$$C_n = D_n \left(\frac{p}{q} - \sum_{k < n} \frac{1}{a_k} \right) = p \prod_{k < n} a_k - q \sum_{k < n} \prod_{\substack{j < n \\ j \neq k}} a_j$$

is a positive integer as well. From $x_n = 1/a_n + x_{n+1}$,

$$C_{n+1} = D_{n+1} x_{n+1} = a_n D_n \left(x_n - \frac{1}{a_n} \right) = a_n C_n - D_n, \quad D_{n+1} = a_n D_n,$$

which is (1). These are the tail variables of Koizumi's Lemma 4 [12, pp. 11–12], up to a common positive factor. Section 4 constructs the same triple with its centred error $E_n = D_n - (a_n - 1)C_n$, which the present section does not need. Rationality is used here and nowhere else below.

2.1 Sparse agreement forces a primitive integral cubic

For $S \subseteq \mathbb{N}$ write $\underline{d}(S) = \liminf_X \#(S \cap [1, X])/X$.

Lemma 2.3 (periodic obstruction principle). *Let $h \geq 1$, let $j_1, \dots, j_L$ be fixed offsets, and suppose that for every sufficiently large n in one residue class modulo h at least one of $n+j_1, \dots, n+j_L$ lies in S . Then $\underline{d}(S) \geq 1/(Lh)$.*

Proof. There are $X/h + O(1)$ relevant starting indices below X , and each element of S serves as a witness for at most L of them. $\square$

Fix $A \in \mathbb{Q}_{>0}$, $B \in \mathbb{Q}$, and suppose for contradiction that

$$P(n) = A n(n+1)(n+2) + B, \quad S = \{n : C_n \neq P(n)\}, \quad \underline{d}(S) = 0.$$

Everything through the end of Section 2.4 runs under that assumption.

Lemma 2.4 (gcd stabilisation and the primitive shape). *Write $G_n = \gcd(C_n, D_n)$. Then $6A$ is a positive integer, G_n divides $6A$ for every n , and G_n is eventually equal to a positive integer g . On the tail where $G_n = g$, put $u_n = C_n/g$, $v_n = D_n/g$ and $Q(n) = P(n)/g$. Then*

$$u_{n+1} = a_n u_n - v_n, \quad v_{n+1} = a_n v_n, \quad \gcd(u_n, v_n) = 1, \quad \gcd(u_n, u_{n+1}) = 1, \quad (2)$$

and there are $m \in \mathbb{Z}_{>0}$ and $c \in \{-1, 1\}$ with

$$Q(n) = \frac{m}{6} n(n+1)(n+2) + c. \quad (3)$$

Proof. Equation (1) gives $G_n \mid G_{n+1}$ and $G_n \mid C_t$ for every $t \geq n$. Since $\underline{d}(S) = 0$, beyond every threshold there is a block of four consecutive indices disjoint from S : otherwise every block of four would meet S and $\underline{d}(S) \geq 1/4$. On such a block $\Delta^3 C_t = \Delta^3 P(t) = 6A$, and G_n divides each of the four values, hence divides $6A$. So $6A$ is a positive integer and the divisibility chain (G_n) is bounded and stabilises at some g .

On the primitive tail, $\gcd(u_n, v_n) = 1$ by construction, and a prime dividing u_n and u_{n+1} would divide $v_n = a_n u_n - u_{n+1}$, which gives $\gcd(u_n, u_{n+1}) = 1$.

The polynomial Q takes an integer value at every integer argument. Indeed, let N clear the denominators of its coefficients, so $NQ \in \mathbb{Z}[X]$ and $Q(n+N) - Q(n) \in \mathbb{Z}$ for every n . A non-integral value at one argument would therefore force an exception at every late argument in one residue class modulo N , and Lemma 2.3 with $L = 1$ would contradict $\underline{d}(S) = 0$. Now $Q(-1) = Q(0) = B/g$, so $c := B/g \in \mathbb{Z}$, and $m := Q(1) - Q(0) = 6A/g$ is a positive integer, which is (3) with $c \in \mathbb{Z}$.

It remains to exclude $c = 0$ and every c with a prime factor. Let ℓ be a prime dividing c , or any prime if $c = 0$. For $n \equiv -1 \pmod{6\ell}$, write $n+1 = 6\ell k$; then

$$\frac{n(n+1)(n+2)}{6} = \ell k(6\ell k - 1)(6\ell k + 1), \quad \frac{(n+1)(n+2)(n+3)}{6} = \ell k(n+2)(n+3),$$

so ℓ divides both $Q(n)$ and $Q(n+1)$. Agreement at both indices would give $\ell \mid \gcd(u_n, u_{n+1}) = 1$. Hence one of $n, n+1$ lies in S for every late $n \equiv -1 \pmod{6\ell}$, and Lemma 2.3 with $L = 2, h = 6\ell$ gives $\underline{d}(S) \geq 1/(12\ell)$. This contradiction leaves $c = \pm 1$. $\square$

Lemma 2.5 (the multiplier supply, and the reducible case). *On the primitive tail, $\gcd(a_n, v_n) = 1$, the multipliers at distinct indices are pairwise coprime, infinitely many of them exceed 1, and the cubic $Q_{m,c}$ of (3) is irreducible over $\mathbb{Q}$.*

Proof. A prime dividing a_n and v_n would divide u_{n+1} and v_{n+1} , against $\gcd(u_{n+1}, v_{n+1}) = 1$; and every earlier multiplier divides every later v_n , so distinct multipliers are coprime. If $a_n = 1$ for all large n then v_n is eventually a positive constant and $u_{n+1} = u_n - v_n$ decreases without bound, against positivity. So infinitely many distinct primes divide late multipliers.

Suppose $Q_{m,c}$ had a rational root. Choose a prime $\ell \nmid 6m$ dividing a late multiplier a_N , large enough that the root reduces modulo ℓ . Then $Q_{m,c}$ vanishes modulo ℓ on a full residue class, while $\ell \mid v_n$ and therefore $\ell \nmid u_n$ for every $n > N$. Agreement at any late index of that class is impossible, and Lemma 2.3 with $L = 1, h = \ell$ gives $\underline{d}(S) \geq 1/\ell$. $\square$

2.2 The square forced by the transport

Lemma 2.6 (square specialisation). *Let $f \in \mathbb{Q}[T]$ be irreducible with root α , and let $H \in \mathbb{Q}[T]$ satisfy $H(\alpha) \neq 0$. If for all but finitely many primes ℓ every root $r \in \mathbb{F}_\ell$ of the reduction of f has $H(r)$ a square in $\mathbb{F}_\ell^\times$, then $H(\alpha)$ is a square in $\mathbb{Q}(\alpha)^\times$.*

Proof. Suppose not. Put $K = \mathbb{Q}(\alpha)$ and choose β with $\beta^2 = H(\alpha)$, so $K(\beta)/K$ is quadratic. Take a finite Galois extension $L/\mathbb{Q}$ containing $K(\beta)$ and $\sigma \in \text{Gal}(L/K)$ restricting non-trivially to $K(\beta)$, so that $\sigma\alpha = \alpha$ and $\sigma\beta = -\beta$. By the Chebotarev density theorem in the form stated by Stevenhagen and Lenstra [10, §3, p. 15], there are infinitely many

unramified rational primes whose Frobenius class in $\text{Gal}(L/\mathbb{Q})$ is that of σ ; only the infinitude is used. Discard the finitely many primes dividing denominators or the discriminant, the prime 2, and those at which β reduces to zero. At a prime of L above such an ℓ with Frobenius σ , the residue field satisfies $\bar{\alpha}^\ell = \bar{\alpha}$ and $\bar{\beta}^\ell = -\bar{\beta}$, so $\bar{\alpha} \in \mathbb{F}_\ell$ while $\bar{\beta} \notin \mathbb{F}_\ell$. Then $H(\bar{\alpha}) = \bar{\beta}^2$ is a nonsquare in $\mathbb{F}_\ell$ at a root of the reduction of f , against the hypothesis. $\square$

The hypothesis is needed at every root and at all but finitely many primes. A favourable finite set of tested primes would not suffice.

Proposition 2.7 (the transport square). *Write $\kappa = m/6$ and $\eta = 6c/m$, so that $Q_{m,c}(n) = \kappa f(n+1)$ for $f(T) = T^3 - T + \eta$. Then $\alpha^2 - 1$ is a square in $\mathbb{Q}(\alpha)^\times$ for a root α of f .*

Proof. Eliminating v_n from (2) gives

$$u_{n+2} = (a_n + a_{n+1})u_{n+1} - a_n^2 u_n. \quad (4)$$

Let ℓ be a prime and k an index with $\ell \mid u_k$. Reading (4) at $n = k-1$ modulo ℓ gives $u_{k+1} \equiv -a_{k-1}^2 u_{k-1}$, hence

$$-u_{k-1}u_{k+1} \equiv (a_{k-1}u_{k-1})^2 \pmod{\ell}, \quad (5)$$

so $-u_{k-1}u_{k+1}$ is a square modulo ℓ .

Let $\ell \nmid 6m$ and let $r \in \mathbb{F}_\ell$ be a root of f . Then $r \neq 0$ and $r \neq \pm 1$, since $f(0) = \eta \neq 0$ and $f(\pm 1) = \eta$. Using $\eta = r - r^3$,

$$f(r-1) = -3r(r-1), \quad f(r+1) = 3r(r+1),$$

so at any index $k \equiv r-1 \pmod{\ell}$ the polynomial values satisfy $Q(k) \equiv 0$, $Q(k-1) = -3\kappa r(r-1)$, $Q(k+1) = 3\kappa r(r+1)$ and

$$-Q(k-1)Q(k+1) = 9\kappa^2 r^2 (r^2 - 1). \quad (6)$$

If $r^2 - 1$ were a nonsquare modulo ℓ , then so would be the right side of (6), because $9\kappa^2 r^2$ is a nonzero square; agreement at all three of $k-1, k, k+1$ would then contradict (5). That prohibition recurs at every index of the class $r-1$ modulo ℓ , and Lemma 2.3 with $L = 3$, $h = \ell$ would give $\underline{d}(S) \geq 1/(3\ell)$. So $r^2 - 1$ is a square in $\mathbb{F}_\ell^\times$ at every root of every good reduction, and Lemma 2.6 applies with $H(T) = T^2 - 1$. $\square$

2.3 The cubic field leaves one scale

Lemma 2.8 (classification of the scales). *Let $m \geq 1$ and $c = \pm 1$, let $\eta = 6c/m$, and suppose $T^3 - T + \eta$ is irreducible over $\mathbb{Q}$ with a root α such that $\alpha^2 - 1$ is a square in $\mathbb{Q}(\alpha)^\times$. Then $m = 12$.*

Proof. Choose $\beta \in \mathbb{Q}(\alpha)$ with $\beta^2 = \alpha^2 - 1$ and put $z = \alpha + \beta$. Then $(\alpha + \beta)(\alpha - \beta) = 1$, so

$$z^{-1} = \alpha - \beta, \quad \alpha = \frac{1}{2}(z + z^{-1}), \quad (7)$$

and z generates $\mathbb{Q}(\alpha)$. Write its minimal polynomial as $Z^3 + bZ^2 + dZ + w$ with $b, d, w \in \mathbb{Q}$ and $w \neq 0$. From $T^3 - T + \eta$,

$$\text{Tr } \alpha = 0, \quad \text{Tr } \alpha^2 = 2, \quad \text{Tr } \alpha^3 = -3\eta. \quad (8)$$

Newton's identities give $\text{Tr } z = -b$ and $\text{Tr } z^{-1} = -d/w$, so the first equation in (8) and (7) give

$$d = -bw. \quad (9)$$

Next $\text{Tr } z^2 = b^2 - 2d = b^2 + 2bw$ and $\text{Tr } z^{-2} = b^2 - 2b/w$, while $\text{Tr}(zz^{-1}) = 3$, so $4 \text{Tr } \alpha^2 = \text{Tr } z^2 + 6 + \text{Tr } z^{-2}$ and the second equation in (8) reads $b^2 + b(w - w^{-1}) = 1$. Multiplying by w factors this exactly:

$$(bw - 1)(b + w) = 0. \quad (10)$$

Similarly $\text{Tr } z^3 = -b^3 - 3b^2w - 3w$ and $\text{Tr } z^{-3} = b^3 - 3b^2/w - 3/w$, while $\text{Tr } z + \text{Tr } z^{-1} = 0$ by (9), so $8 \text{Tr } \alpha^3 = \text{Tr } z^3 + \text{Tr } z^{-3}$ and the third equation in (8) gives

$$\eta = \frac{(b^2 + 1)(w + w^{-1})}{8}. \quad (11)$$

By (10), either $b = -w$ or $b = w^{-1}$, and (11) becomes

$$\eta = \frac{(w^2 + 1)^2}{8w} \quad \text{or} \quad \eta = \frac{(w^2 + 1)^2}{8w^3}.$$

Write $w = r/s$ with $r \in \mathbb{Z} \setminus \{0\}$, $s \in \mathbb{Z}_{>0}$ and $\gcd(r, s) = 1$. Since $\eta = 6c/m$,

$$m = \frac{48c rs^3}{(r^2 + s^2)^2} \quad \text{or} \quad m = \frac{48c r^3 s}{(r^2 + s^2)^2}. \quad (12)$$

Now $\gcd(r^2 + s^2, rs) = 1$, so integrality of m and $|c| = 1$ force $(r^2 + s^2)^2 \mid 48$, hence $r^2 + s^2 \in \{1, 2, 4\}$. With $r \neq 0$ and $s \geq 1$ the value 1 is too small, and 4 is not a sum of two nonzero squares, so $r^2 + s^2 = 2$ and $|r| = s = 1$. Then (12) gives $m = 12cr$, and $m > 0$ gives $m = 12$. $\square$

The square condition alone does not exclude the two survivors

$$Q_{12, \pm 1}(n) = 2n(n+1)(n+2) \pm 1. \quad (13)$$

The last step uses the successor coupling that the square condition forgets.

2.4 Both survivors fail modulo seven

Lemma 2.9 (the two forbidden words). *For $Q_{12,1}$, every sufficiently late four-index window beginning at $n \equiv 0 \pmod{7}$ contains an exceptional index. For $Q_{12,-1}$, the same holds for windows beginning at $n \equiv 1 \pmod{7}$. In either case, $\underline{d}(S) \geq 1/7$. No phase-free four-index obstruction is asserted.*

Proof. Reduce modulo 7. For $c = 1$ the values of $Q_{12,1}$ at $n = 0, 1, 2, 3$ are 1, 13, 49, 121, that is

$$(1, 6, 0, 2), \quad (14)$$

and for $c = -1$ the values at $n = 1, 2, 3, 4$ are 11, 47, 119, 239, that is

$$(4, 5, 0, 1). \quad (15)$$

Each word recurs in n with period 7.

Let $(c_0, c_1, 0, c_3)$ be the residues of u at four consecutive indices of the primitive tail and (d_0, d_1, d_2, d_3) those of v , so that $c_{i+1} = a_i c_i - d_i$ and $d_{i+1} = a_i d_i$ modulo 7. The middle zero gives $a_1 c_1 = d_1$, then $d_2 = a_1 d_1$ and $c_3 = -d_2$, so

$$d_1^2 = c_1 a_1 d_1 = c_1 d_2 = -c_1 c_3. \quad (16)$$

For (14) this is $-6 \cdot 2 = 2$, and for (15) it is $-5 \cdot 1 = 2$. The square roots of 2 modulo 7 are 3 and 4, so $d_1 \in \{3, 4\}$ in both cases.

The first step gives $a_0 = (c_1 + d_0)/c_0$ and hence $d_1 = d_0(d_0 + c_1)/c_0$, which is legitimate because $c_0 \in \{1, 4\}$ is invertible. For (14) this is $d_1 = d_0(d_0 + 6)$, with values

$$(0, 0, 2, 6, 5, 6, 2) \quad \text{at } d_0 = 0, 1, \dots, 6,$$

and for (15) it is $d_1 = 2d_0(d_0 + 5)$, with values

$$(0, 5, 0, 6, 2, 2, 6).$$

Both images are $\{0, 2, 5, 6\}$, which misses $\{3, 4\}$. All seven initial denominator residues have been displayed, so the calculation is exhaustive and rests on no unreported search.

Hence every sufficiently late window $[7k, 7k + 3]$ for the plus profile, and every sufficiently late window $[1 + 7k, 4 + 7k]$ for the minus profile, contains an element of S . Windows within each family are pairwise disjoint. There are $X/7 + O(1)$ such windows contained in $[0, X]$, so choosing one exceptional index in each gives $\#(S \cap [0, X]) \geq X/7 - O(1)$ and therefore $\underline{d}(S) \geq 1/7$. This stronger local bound is not promoted to a uniform lower bound for arbitrary cubic profiles. $\square$

Proof of Theorem 2.2. Assume $\underline{d}(S) = 0$. Lemmas 2.4 and 2.5 put the primitive tail in the shape (3) with $c = \pm 1$ and $Q_{m,c}$ irreducible. Each excluded branch already contradicts the assumption by producing a positive, branch-dependent lower density, namely $1/(12\ell)$ or $1/\ell$ for the relevant prime ℓ . Proposition 2.7 and Lemma 2.8 force $m = 12$, and Lemma 2.9 then gives $\underline{d}(S) \geq 1/7$, the final contradiction. $\square$

2.5 From a regular rate to an exact cubic

Lemma 2.10 (integer extraction at a regular rate). *Let $\lambda > 1$ and let C_n be positive integers with*

$$\frac{C_{n+1}}{C_n} = 1 + \frac{\lambda}{n} + o(n^{-\lambda}). \quad (17)$$

Then λ is an integer $d \geq 2$, and there are $A \in \mathbb{Q}_{>0}$ and $B \in \mathbb{Q}$ with $C_n = A n(n+1) \cdots (n+d-1) + B$ for all large n .

Proof. Put $F_n = \Gamma(n + \lambda)/\Gamma(n)$, so that $F_{n+1}/F_n = 1 + \lambda/n$ exactly and $F_n \sim n^\lambda$ by the gamma-ratio asymptotic [20, 5.11.12]. Write $\varepsilon_n = o(n^{-\lambda})$ for the error in (17) and $z_n = C_n/F_n$, so that $z_{n+1}/z_n = 1 + \varepsilon_n/(1 + \lambda/n)$. Since $\lambda > 1$ the errors are absolutely summable, so the product converges and $z_n \rightarrow K$ for some $K > 0$, with $z_n - K = o(n^{1-\lambda})$. To see the stated error, put $\eta_n = \sup_{k \geq n} k^\lambda |\varepsilon_k| \rightarrow 0$; the tail of the logarithmic product has absolute value at most $O(\eta_n \sum_{k \geq n} k^{-\lambda}) = o(n^{1-\lambda})$. The factors are positive eventually, so the limiting product is nonzero. Hence

$$\delta_n := C_n - KF_n = F_n(z_n - K) = o(n). \quad (18)$$

The recurrence gives $\delta_{n+1} - \delta_n = (\lambda/n)\delta_n + \varepsilon_n C_n$, in which the first term is $o(1)$ by (18) and the second is $o(1)$ because $C_n = O(n^\lambda)$. So $\Delta \delta_n \rightarrow 0$, and therefore $\Delta^j \delta_n \rightarrow 0$ for every $j \geq 1$.

For F , the identity $\Gamma(n+1+\lambda)/\Gamma(n+1) - \Gamma(n+\lambda)/\Gamma(n) = \lambda \Gamma(n+\lambda)/\Gamma(n+1)$ iterates to

$$\Delta^j F_n = \lambda(\lambda-1) \cdots (\lambda-j+1) \frac{\Gamma(n+\lambda)}{\Gamma(n+j)},$$

whose right side is $O(n^{\lambda-j})$. Choose an integer $j > \lambda$; then $\Delta^j F_n \rightarrow 0$, so $\Delta^j C_n \rightarrow 0$. These are integers, so $\Delta^j C_n = 0$ for all large n , and C_n agrees eventually with a polynomial with rational coefficients. Its growth $C_n \asymp n^\lambda$ forces the degree to be λ , so $\lambda = d$ is an integer, and $d \geq 2$ because $\lambda > 1$. For that integer $F_n = n(n+1) \cdots (n+d-1)$ exactly, so (18) says that the difference of two polynomials of degree d is $o(n)$, hence constant. The leading coefficient of the rational polynomial is K , so $K \in \mathbb{Q}_{>0}$; the constant difference is rational as well. This gives the stated shape with $A = K$ and $B \in \mathbb{Q}$. $\square$

The precision in (17) carries weight. The positive integers $C_n = n(n+1)(n+2) + (-1)^n$ satisfy $C_{n+1}/C_n = 1 + 3/n + O(n^{-3})$ and are not eventually polynomial, so the little- o remainder cannot in general be replaced by a big- O remainder at the same exponent. That countermodel is a scalar sequence rather than an exact orbit, and it bears on Lemma 2.10 alone.

Lemma 2.11 (the tail ratio reads the growth defect). *Let a_n be strictly increasing positive integers with $a_{n+1}/a_n^2 \rightarrow 1$ and $\sum_n 1/a_n$ rational, and let (C_n, D_n) be the canonical integer tail. Write $\gamma_n = a_n^2/a_{n+1} - 1$. Then*

$$\frac{C_{n+1}}{C_n} = 1 + \gamma_n + O(a_n^{-1}), \quad a_n \geq \exp(c 2^n) \text{ eventually for some } c > 0.$$

Proof. Since $a_{n+1}/a_n^2 \rightarrow 1$ and $a_n \rightarrow \infty$, there is an N with $a_{n+1} \geq a_n^2/2 \geq 2a_n$ for $n \geq N$. Iterating $a_{n+1} \geq a_n^2/2$ from an index where $a_n > 2$ gives $a_n \geq \exp(c 2^n)$ eventually. The same doubling gives, for $m > N$,

$$\frac{1}{a_m} \leq x_m = \sum_{k \geq m} \frac{1}{a_k} \leq \frac{2}{a_m},$$

so $x_m = (1 + \rho_m)/a_m$ with $0 \leq \rho_m = a_m x_{m+1} \leq 2a_m/a_{m+1} \leq 4/a_m$, using $a_{m+1} \geq a_m^2/2$. Hence

$$\frac{C_{n+1}}{C_n} = \frac{a_n D_n x_{n+1}}{D_n x_n} = a_n \cdot \frac{(1 + \rho_{n+1})/a_{n+1}}{(1 + \rho_n)/a_n} = \frac{a_n^2}{a_{n+1}} \cdot \frac{1 + \rho_{n+1}}{1 + \rho_n}.$$

The last factor is $1 + O(1/a_n)$ and a_n^2/a_{n+1} is bounded, so the product is $a_n^2/a_{n+1} + O(a_n^{-1})$. $\square$

Equation (20) of Section 6 records the same comparison in the centred coordinates, where it follows from the kernel-checked defect identity of Theorem 5.1.

Proof of Theorem 2.1. The hypothesis $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$ gives $a_{n+1}/a_n^2 \rightarrow 1$. Suppose $\sum_n 1/a_n$ were rational and pass to the canonical integer tail, so that (a, C, D) satisfies (1) with every C_n and D_n a positive integer. By Lemma 2.11 and $a_n \geq \exp(c2^n)$, the error term $O(a_n^{-1})$ is $o(n^{-3})$, so

$$\frac{C_{n+1}}{C_n} = 1 + \frac{3}{n} + o(n^{-3}).$$

Lemma 2.10 at $\lambda = 3$ gives $C_n = A n(n+1)(n+2) + B$ for all large n , with $A \in \mathbb{Q}_{>0}$ and $B \in \mathbb{Q}$. That contradicts Theorem 2.2. $\square$

The composition handles every rational normalisation of the tail. It does not assume that the primitive leading coefficient is 1, or 2, or bounded in advance.

This is an ordinary proof. The finite claims include the two words (14) and (15), the image $\{0, 2, 5, 6\}$ modulo 7, the divisor condition $(r^2 + s^2)^2 \mid 48$, and the scalar sequence $n(n+1)(n+2) + (-1)^n$. The public checkpoint also contains formalised polynomial extraction, primitive cubic normalisation and finite transport results, including the primitive-shape theorem. These are components, not an assembled proof of Theorem 2.1. No declaration of that complete irrationality implication has been identified in the supplied source index. Lemma 2.6 uses the classical finite-Galois Chebotarev theorem; its use must be justified by the number-field argument, not by finite residue computations.

Position. Theorem 2.1 carries no rationality hypothesis and no state hypothesis, and it is the principal irrationality theorem proved in this record. Its rate class lies inside the growth hypothesis of Problem 1.1, because $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$ forces $a_{n+1}/a_n^2 \rightarrow 1$. On that class the assertion of Problem 1.1 therefore holds, by exclusion of a rational reciprocal sum, and the recurrence is never exhibited; Sylvester's own sequence, whose ratio a_n^2/a_{n+1} is $1 + O(1/a_n)$, lies outside the class. The rate $1 + 3/n$ sits beyond the hypotheses recorded in Section 3. The product weight P_n/a_n grows like n^3 at this rate, so $(P_n/a_n)(a_n^2/a_{n+1} - 1)$ tends to infinity and Koizumi's Corollary 4(1) does not apply; the rate also fails his $1 + o(1/n)$ condition, and $\sum_n \gamma_n$ diverges, so Duverney's convergent-defect hypothesis does not apply. Whether the LCM-weighted criteria apply on this class is not decided here, since the ratio of the two weights is the overlap factor. Theorem 2.2 uses positive integer a_n, C_n, D_n and the exact recurrences (1), but does not require near-quadratic growth. The later sufficient criteria are not asserted to subsume the cubic-rate argument.

The display defining the canonical integer tail gives $C_n = q(\prod_{k < n} a_k \cdot \sum_m 1/a_m - \sum_{k < n} \prod_{j < n, j \neq k} a_j)$, a linear form in the reciprocal sum with integer coefficients. Approximation criteria for irrationality use nonzero forms of this kind that tend to zero. At the cubic rate the integers C_n grow: Lemma 2.11 gives $C_{n+1}/C_n = 1 + 3/n + o(n^{-3})$,

so C_n is a positive multiple of $n(n+1)(n+2)$ up to a factor tending to one. The proof above uses this rate at precision $o(n^{-3})$: Lemma 2.10 turns it into the exact profile $C_n = A n(n+1)(n+2) + B$, which Theorem 2.2 excludes.

3 Prior work, and the status of the note

Classical sufficient conditions and the pseudo-greedy formulation. The rational-tail method predates these coordinates. Erdős and Straus [15, Theorem 2.1, pp. 85–86] use eventual integer remainders under a small-numerator hypothesis. For polynomial Cantor series, Hančl and Tijdeman decompose the numerator into finitely many shifted products; rationality is characterised by the vanishing of the resulting polynomial sum [16, Theorem 2.2 and its proof, pp. 39–40]. We use that polynomial theorem as methodological context, not as an unrestricted theorem about arrays of signed integer coefficients. The finite-to-infinite rearrangement requires boundary control, which is available in the polynomial setting. Neither cited result supplies the bounded-negative first-crossing argument proved below.

Several classical results give sufficient conditions directly on the sequence (a_n) . Their precise hypotheses must be compared separately with the bounded-defect, summability and prescribed-rate conditions here. Duverney imposes a different sufficient analytic hypothesis; the Erdős–Straus one, read through the dictionary of the next paragraph, is implied by nonnegativity of E , the hypothesis of Theorem 8.1, so that criterion already covers the descent of Section 8 (see [12, Cor. 4(1) and Prop. 1(1), pp. 14–15]). Erdős and Straus proved that if $\lim a_n/a_{n-1}^2 = 1$, the reciprocal sum is rational, and a_n does not satisfy the recurrence, then

$$\limsup_{n \rightarrow \infty} \frac{[a_1, \dots, a_n]}{a_{n+1}} \left(\frac{a_{n+1}^2}{a_{n+2}} - 1 \right) > 0,$$

where $[a_1, \dots, a_n]$ is the least common multiple [1, Theorem 3, p. 132]. This is the indexing in the original theorem. Erdős’s 1988 survey, followed by the supplied catalogue summary, instead prints $a_n^2/a_{n+1} - 1$ in the second factor while retaining the same prefix-LCM quotient; that displayed summary is off by one and is not followed here [6, p. 105][13]. Koizumi records the convenient sufficient rate $a_n^2/a_{n+1} = 1 + o(1/n)$ under which the Erdős–Straus criterion settles the problem [12, Remark 3, p. 16]. Tijdeman and Yuan give a criterion of the same kind for Ahmes series with positive integer numerators, weighted by the least common multiple of the earlier denominators [4].

Duverney proved a conditional signed form of the problem itself: for signs $\epsilon_n \in \{-1, 1\}$, if $\sum_{n \geq 0} (a_{n+1}/a_n^2 - 1)$ converges, then the reciprocal sum with numerators ϵ_n is rational if and only if

$$a_{n+1} = a_n^2 - (\epsilon_{n+1}/\epsilon_n)a_n + \epsilon_{n+2}/\epsilon_{n+1}$$

for all large n [2, Corollary 3.2, p. 287]. Absolute convergence is a stronger sufficient specialisation, not the hypothesis printed in that corollary. The all-positive specialisation is the form relevant here.

A different quantitative question is treated by Duverney, Kurosawa and Shiokawa [17, Theorem 1, author-version p. 2; proof in Section 3]. For rational $x_n > 1$, their result

assumes eventual $x_{n+1} \geq x_n^2$ and control of the accumulated denominators of x_{n+1}/x_n^2 ; it computes the irrationality exponent of a signed reciprocal series. We use it only as a restricted comparison. In particular, a Sylvester tail and the cubic rate considered here approach quadratic growth from the other side.

Badea's positive-term criterion is adjacent but different: for positive rational terms b_n/a_n , eventual strict inequality

$$a_{n+1} > \frac{b_{n+1}}{b_n} a_n^2 - \frac{b_{n+1}}{b_n} a_n + 1$$

forces irrationality, while rationality under the corresponding non-strict inequality forces eventual equality [3, Theorem A, p. 313; Cor. 2.2, p. 316]. For $b_n = 1$, the non-strict inequality is compatible with the familiar nonnegative-error descent after passage to the rational tail. It does not cover arbitrary signed centred errors. The general positive-coefficient criterion is not identified as a checked theorem in the supplied evidence.

Koizumi's work on doubly exponential sequences [12] bears directly on the state system used below, and supplies the coordinates in which several of the statements of this note are most naturally read. For a rational $r = p/q$ with pseudo-greedy expansion (a_n) , remainders (x_n) and gap sequence $\varepsilon_n = x_n^{-1} + 1 - a_n$, his Lemma 4 [12, pp. 11–12] produces integers $c_n > 0$, d_n and e_n with $x_n = c_n/d_n$ and $\varepsilon_n = e_n/c_n$, satisfying

$$a_n = \frac{d_n - e_n}{c_n} + 1, \quad c_{n+1} = c_n - e_n, \quad d_{n+1} = a_n d_n,$$

and the proof of that lemma also gives $c_{n+1} = a_n c_n - d_n$. Under the dictionary $(C_n, D_n, E_n) = (c_n, d_n, e_n)$ these are exactly the objects of Section 4: the first relation is the centring map ctr rewritten as $e_n = d_n - (a_n - 1)c_n$, the second is Proposition 4.1, the third is the denominator update, and the fourth is the tail update. The lower-case e_n of Section 4 is $-E_n$, so it is the negative of Koizumi's e_n at an index where the centred state is negative.

His Theorem 3 [12, pp. 12–13] shows that his Conjecture 1 [12, p. 4], that for a positive rational r whose gap sequence satisfies $\varepsilon_n \rightarrow 0$ one has $\varepsilon_n = 0$ for all large n , is equivalent to an affirmative answer to the question of Erdős and Graham recorded as his Question 1 [12, p. 3], which is Problem 1.1 above. Two of the implications proved below have prior art in these canonical coordinates: his Lemma 3, that $\varepsilon_n = 0$ forces $\varepsilon_{n+1} = 0$, is the absorption of Theorem 5.8, and his Proposition 1(2), that $\varepsilon_n \geq 0$ for all large n forces $\varepsilon_n = 0$ for all large n , is the descent of Theorem 8.1. These appear in canonical coordinates in [12, Lemma 3, p. 10; Prop. 1(2), p. 14]. Koizumi attributes Proposition 1(2) to Badea. Its contrapositive says that a counterexample must have negative errors infinitely often; it assumes the sign is eventually nonnegative, while Theorem 12.1 below assumes only that the negative part is eventually bounded, which admits sign changes of bounded depth at every scale.

The growth hypothesis in Problem 1.1 is calibrated by two facts about Sylvester's sequence. It satisfies $a_n \approx c_0^{2^n}$ with $c_0 = 1.2640847\dots$, and shifting it further produces sequences with $a_n \approx C^{2^n}$ for arbitrarily large C whose reciprocals still sum to a rational number [7, arXiv v4, p. 2]. The classical sufficient condition for irrationality, $\lim_n a_n^{1/2^n} = \infty$, is therefore sharp. Kovač and Tao identify that condition as folklore [7, arXiv v4, p. 2]; they attribute the sharpness observation to Erdős (1975). A

sequence with $a_n/a_{n-1}^2 \rightarrow 1$ has $a_n^{1/2^n}$ convergent, so that criterion says nothing about the sequences of Problem 1.1, and rationality is genuinely possible there. Sylvester's sequence is A000058 in the OEIS and its shifts A129871. For the recent literature on irrationality of Ahmes series we refer to Kovač and Tao [7], who resolve several problems of Erdős and Graham drawn from the same two sources cited above, and whose introduction gives a sample of the intermediate work, including Sándor (1984) and Badea (1987). They do not treat Problem #243; the rigidity conclusion asked for there is not among their results.

Crmarčić and Kovač [18, Theorems 1–2] address the neighbouring Problem #270, not Problem #243. Allowing integer $f(n) \rightarrow \infty$ in $\sum_n (\prod_{j=1}^{f(n)} (n+j))^{-1}$ gives every positive real value; imposing nondecreasing f gives a measure-zero value set. The latter assertion does not rule out individual rational values. Their extension of Makeyev's subsum argument explains why decay alone need not force irrationality when the summands remain freely selectable. The present exact denominator recurrence imposes additional compatibility, so neither direction is an implication between their theorem and ours.

The *Formal Conjectures* collection contains a mathematically equivalent unproved declaration, up to its zero-based indexing [14]. Its summand is $\mathbb{Q}$ -valued, so Lean's Summable hypothesis asserts the existence of a sum in $\mathbb{Q}$; the finite indexing shift changes that sum only by a rational prefix. The declaration therefore does encode the rationality premise, but its proof is sorry. Its role here is statement-level prior art; it supplies no proof authority, and the development in this note is independent of it. The checked results in this project include the state implications and the specified original-coordinate transfer theorems; they are not confined to the state system. Earlier formal work on the remainder method should also be distinguished from a solution of this problem: Koutsoukou-Argraki and Li's Archive of Formal Proofs entry [19] records an Isabelle/HOL formalisation of Erdős–Straus (1974), Theorem 2.1, Corollary 2.10 and Theorem 3.1. That is formal prior art for those classical criteria, not an existing formal proof of Erdős #243.

Results and proof status. The cubic-rate theorem (Theorem 2.1) is an ordinary irrationality proof with formalised components. The bounded-negative theorem (Theorem 12.1) and the scalar finite-mass theorem (Theorem 13.1) have checked declarations. The checked canonical product- and LCM-defect theorems include their analytic transfer. The complete weighted-record statement has a proof source in the separate release; that source is not part of the main-repository build documented here. For the further record criteria, each evidence paragraph identifies the ordinary argument and the checked lemma it uses. In particular, Lemma 7.7 below proves the canonical odd-prime-power supply on the page; its new proof is not included in the inherited Lean receipt. The static-model conclusions in Section 14 are likewise ordinary results.

The additional bound or summability assumption is not supplied by the problem's hypotheses. The record-excess equivalence therefore identifies an obligation; it does not discharge it. In the stable-gcd case the record-increment theorem gives a positive lower bound for the log-log coefficient, not necessarily an infinite coefficient. All nontermination conclusions retain the failure of eventual Sylvester behaviour as a hypothesis.

The proof of bounded-negative rigidity can be read independently of the cubic argument: exact tail updates, absorption, gcd stabilisation, persistent coprimality and a first crossing of a Chinese-remainder block. The finite-mass proof instead uses a bounded infinite product and descent of positive integers.

The remaining exclusions sharpen this boundary. By Theorem 5.8, a counterexample has $E_n \neq 0$ for every large n . By Theorem 8.1 it cannot have E eventually nonnegative. If its tail is all negative, Theorems 9.2 and 10.1 exclude, respectively, constant magnitude and periodic magnitude in the stated regime $e_n < a_n$ with positive drift; they are not a classification of a mixed-sign tail. Under normalised vanishing, Theorems 12.1 and 13.1 show that the negative part is neither bounded nor of finite normalised mass. What survives is stated in Section 14.

Structure

There are three independent reading routes. For the cubic theorem, read Section 2, following the chain: integer tail, polynomial extraction, primitive cubic, transport square, scale classification, modulo-seven contradiction. For bounded-negative rigidity, read Sections 4, 5, 11 and 12; neither the cubic field nor the energy criterion is needed. For record criteria, read Sections 6 and 7, keeping LCM and primitive records distinct.

Section 13 is the short scalar product argument. Sections 9–10 retain earlier special cases and their formal sources. Section 14 separates necessary conditions on an actual reciprocal tail from statements about static avoidance models; it now answers the draft's two auxiliary barrier questions under their stated hypotheses. Appendix A maps formal sources, and Appendix B explains the finite residue reduction. Pinned source links identify proof locations, not an assertion that every surrounding paragraph has been formalised.

4 The centred integer state

The three maps defined next carry out the two steps announced at the end of Section 1. The first two clear denominators along the sequence, so that a rational reciprocal sum leaves a sequence of integers; the third subtracts from that integer the value it takes on Sylvester's sequence, leaving one number whose vanishing is the whole question. All three are maps between integer tuples and carry no hypothesis of any kind. The recurrences they encode are Koizumi's [12, Lemma 4, pp. 11–12]; the defect, scaling and two-step identities below are algebraic consequences of them in the conventions of the formal sources.

For $a, D, C \in \mathbb{Z}$ define

$$\text{den}(a, D) = aD, \quad \text{tail}(a, D, C) = aC - D, \quad \text{ctr}(a, D, C) = D - (a - 1)C,$$

the [denominator update](#), the [tail update](#), and the [centred state](#). Given a sequence (a_n) of integers, whose terms we call the *multipliers*, we write $D_{n+1} = \text{den}(a_n, D_n)$ for the *denominator state*, $C_{n+1} = \text{tail}(a_n, D_n, C_n)$ for the *tail state*, and $E_n = \text{ctr}(a_n, D_n, C_n)$ for the *centred state*, also called the *error*. The word *centred* records what the third map does: it measures D_n against $(a_n - 1)C_n$, and $D_n = (a_n - 1)C_n$ holds at every index on Sylvester's

sequence (Example 4.2), so E is a deviation from that sequence rather than a size. At an index where $E_n < 0$ we write $e_n = -E_n > 0$ and call it the *magnitude* of the error there.

Proposition 4.1 (update law). $\text{tail}(a, D, C) = C - \text{ctr}(a, D, C)$; that is, $C_{n+1} = C_n - E_n$.

Proof. $C - (D - (a - 1)C) = aC - D$. □

Formalised as the [update law](#). The state therefore moves by exactly the error at each step, which is what makes the sign of E decisive: where E is positive the state falls, where E is negative it rises, and where E vanishes it is stationary.

Example 4.2 (the Sylvester orbit). Take $a_n = 2, 3, 7, 43, 1807, \dots$ with $a_{n+1} = a_n^2 - a_n + 1$, and start the state at $D_0 = C_0 = 1$. The two updates give

n	a_n	D_n	C_n	E_n
0	2	1	1	0
1	3	2	1	0
2	7	6	1	0
3	43	42	1	0
4	1807	1806	1	0

and $D_n = a_n - 1$ with $C_n = 1$ at every index: if $D_n = a_n - 1$ and $C_n = 1$ then $C_{n+1} = a_n - (a_n - 1) = 1$ and $D_{n+1} = a_n(a_n - 1) = a_{n+1} - 1$. Hence $E_n = D_n - (a_n - 1)C_n = 0$ throughout and the tail state never moves, which is Proposition 4.1 in the stationary case.

The canonical reciprocal-tail interpretation. Let $T_n = \sum_{k \geq n} 1/a_k$ and suppose $T_0 \in \mathbb{Q}$. Put D_0 equal to a common denominator and $D_n = D_0 a_0 \cdots a_{n-1}$, and set $C_n = D_n T_n$. Then $C_n \in \mathbb{Z}$ for every n , and from $T_n = 1/a_n + T_{n+1}$ one gets $C_{n+1} = a_n C_n - D_n$, which is the tail update. On Sylvester's sequence $T_n = 1/(a_n - 1)$ exactly, so $D_n = (a_n - 1)C_n$ and $E_n = 0$: the centred state measures deviation from the Sylvester tail identity, and it vanishes identically on the Sylvester orbit.

The elementary identities in `ReciprocalTailRigidity.lean` are stated for an abstract integer system. Their canonical reciprocal-tail interpretation is supplied separately by the checked `PaperCompleteR7` modules, including [the product-defect transfer](#). The natural-state identities used in this passage include the [tail realisation](#), the [denominator realisation](#), and the [signed update](#). Keeping these levels separate makes the hypotheses of each implication visible without suggesting that the analytic interpretation is absent.

The series itself is reached by three further declarations, which were absent from earlier revisions of this note. Write C_n/D_n for the real ratio carried by the natural state, the [tail ratio](#). Given $a_n > 0$, $D_n > 0$, $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$, one exact step removes $1/a_n$ from that ratio [one-step reciprocal identity](#), finite iteration telescopes to a partial sum plus the endpoint ratio [finite telescoping](#), and convergence of the endpoint ratio to zero identifies the infinite sum [the series realisation](#): $\sum_n 1/a_n$ then has

sum C_0/D_0 . These three carry no growth hypothesis, no centring hypothesis and no rationality hypothesis, and they exclude nothing. They close the loop between the integer system and the series that motivates it. These three declarations concern state-to-series realisation. The opposite direction, construction from a rational reciprocal sum, is an ordinary calculation here and is also supplied by the separate canonical declarations in PaperCompleteR7. It must not be declared unchecked merely because it is absent from this abstract-state module.

Proposition 4.3 (scale equivariance). *For every $s, a, D, C \in \mathbb{Z}$,*

$$\text{den}(a, sD) = s \text{ den}(a, D), \quad \text{tail}(a, sD, sC) = s \text{ tail}(a, D, C),$$

$$\text{ctr}(a, sD, sC) = s \text{ ctr}(a, D, C).$$

Formalised as the [denominator equivariance](#), the [tail equivariance](#), and the [centred equivariance](#). The system depends only on the ray through (D, C) . This is used twice below: once to normalise the finite search of Appendix B, and once, in Theorem 10.1, as the induction step that removes common scale.

5 The defect identity and its two consequences

Section 4 connects the Sylvester recurrence to the error in one direction only: on Sylvester's sequence the error vanishes. The converse is what the problem needs, and it comes from a single identity, which expresses the deviation of a_{n+1} from the Sylvester successor, multiplied by the tail state C_{n+1} , as a combination of the errors at n and at $n+1$, the first weighted by a_n^2 . Its two consequences are Theorem 5.7, in which an eventually vanishing error forces the recurrence provided the tail state is eventually nonzero, and Theorem 5.8, in which, under strict centring (the condition $|E_n| < C_n$ at every index), one vanishing error propagates to every later index.

Write $\text{def}(a, a') = a' - \text{syl}(a)$ for the deviation of the next term from the Sylvester successor, the [Sylvester defect](#).

Theorem 5.1 (defect identity). *For all $a, a', D, C \in \mathbb{Z}$,*

$$\text{def}(a, a') \cdot \text{tail}(a, D, C) = a^2 \text{ ctr}(a, D, C) - \text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)),$$

that is, $\Delta_n C_{n+1} = a_n^2 E_n - E_{n+1}$.

Proof. A direct expansion: both sides equal $a' a C - a' D - a^3 C + a^2 D + a^2 C - a D - a C + D$. $\square$

Formalised as the [defect identity](#). It is a polynomial identity in four indeterminates, with no hypothesis of any kind, and it carries the whole argument twice over: once for rigidity, and once for absorption.

Example 5.2 (one defect and the error it creates). Continue Example 4.2 but replace $a_3 = 43$ by $a_3 = 44$. The states $D_3 = 42$ and $C_3 = 1$ are unchanged, since they depend only on a_0, a_1, a_2 , and $E_2 = 0$ still. The defect is $\Delta_2 = a_3 - \text{syl}(a_2) = 44 - 43 = 1$, and $E_3 = D_3 - (a_3 - 1)C_3 = 42 - 43 = -1$, so the identity reads

$$\Delta_2 C_3 = 1 \cdot 1 = 1 = 49 \cdot 0 - (-1) = a_2^2 E_2 - E_3.$$

By Proposition 4.1 the tail state then rises, $C_4 = C_3 - E_3 = 2$. A single unit of defect at one index has produced a negative error of magnitude 1 at the next, and the state has started to climb.

Theorem 5.3 (two vanishing errors force the step). *Let $a, a', D, C \in \mathbb{Z}$ with $\text{tail}(a, D, C) \neq 0$. If $\text{ctr}(a, D, C) = 0$ and $\text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)) = 0$, then $a' = \text{syl}(a)$.*

Proof. By Theorem 5.1 the product $\text{def}(a, a') \cdot \text{tail}(a, D, C)$ is zero, and the second factor is nonzero by hypothesis. $\square$

Formalised as the **local rigidity step**. The hypothesis $C_{n+1} \neq 0$ is not removable: at $C_{n+1} = 0$ the identity gives no information about a' .

5.1 Two-step cancellation and feedback

The one-step state equations hide a second-order constraint once the raw numerator is reduced at two consecutive stages. Write u, u_1, u_2 for the primitive numerator coordinates, v, v_1 for the corresponding denominator coordinates, and h, h_1 for the two common factors removed in reduction. If the two steps have multipliers a, a_1 , then the exact hypotheses are

$$hu_1 + v = au, \quad hv_1 = av, \quad h_1u_2 + v_1 = a_1u_1.$$

Theorem 5.4 (two-step primitive recurrence). *Let $a, a_1, u, u_1, u_2, v, v_1, h, h_1$ be integers satisfying*

$$hu_1 + v = au, \quad hv_1 = av, \quad h_1u_2 + v_1 = a_1u_1.$$

Then $a^2u + hh_1u_2 = h(a + a_1)u_1$.

Proof. Multiply the first equation by a , replace av using the second equation, and then replace $h_1u_2 + v_1$ using the third. The remaining terms factor as the displayed right-hand side. $\square$

This is the **two-step primitive recurrence**. The hard point is keeping both changing cancellation factors visible. It is an elimination identity for two exact steps, not a new assumption. It remains non-autonomous and cannot by itself force $a_1 = a^2 - a + 1$; Lemma 5.6 specifies what it forgets modulo a fixed old denominator.

There is a complementary integer identity when the relevant step is cancellation-free. Put $q = ap - p_1$, and suppose the next primitive coordinate satisfies

$$p_2 + a^2p = (a + a_1)p_1.$$

Then

Theorem 5.5 (cancellation-free curvature identity). *Let a, a_1, p, p_1, p_2, q be integers with $q = ap - p_1$ and $p_2 + a^2p = (a + a_1)p_1$. Then*

$$q^2 + (pp_2 - p_1^2) = (a_1 - a)pp_1.$$

The square q^2 is the local remainder, while $pp_2 - p_1^2$ is a discrete curvature term. The right side records the change in multiplier, so the identity supplies a diagnostic for an anti-shadowing argument without asserting a sign or a global monotonicity. It is checked as [the cancellation-free curvature square](#); no global anti-shadowing conclusion is claimed.

Finally, the reduced denominator transport has a precise realisability condition. Let e, e_1 be centred errors and let Δ be the Sylvester defect. If

$$hu_1 = u - e, \quad he_1 = a^2e - \Delta(u - e),$$

then the transport equality

$$h((a_1 - 1)u_1 + e_1) = a((a - 1)u + e)$$

has mismatch which factors through $u - e$. Consequently, when $u - e \neq 0$,

$$h((a_1 - 1)u_1 + e_1) = a((a - 1)u + e) \iff a_1 = a^2 - a + 1 + \Delta.$$

This is the exact [feedback transport equivalence](#), obtained from the factored defect at [feedback transport defect](#). The nonzero-raw-numerator hypothesis is essential: if $u - e = 0$, the factorised mismatch cannot identify a_1 .

The identities hold under the local hypotheses displayed in their statements. It is their proposed global applications, not the identities, that would need additional information, such as control of cancellation or a sign estimate for the curvature. No such information is deduced here for an arbitrary rational near-quadratic tail.

Lemma 5.6 (saturation modulo an old denominator). *Let $M \geq 2$. Any finite word $r_0, \dots, r_k$ of units modulo M is compatible with the cancellation-free recurrences modulo M , with all reduced denominator residues equal to zero. Consequently the eliminated two-step identity alone imposes no further restriction on such unit words when the multiplier residues are free.*

Proof. Set $v_i \equiv 0 \pmod{M}$ and choose $a_i \equiv r_{i+1}r_i^{-1} \pmod{M}$. Then $r_{i+1} + v_i \equiv a_i r_i$ and $v_{i+1} \equiv a_i v_i$. Eliminating v_i gives the two-step identity automatically. $\square$

This is finite residue compatibility, not existence of a positive integer orbit, much less a canonical rational tail with near-quadratic growth. For $M \mid v_T$ on a clean tail, the actual u_n are indeed units modulo M . Useful further restrictions must therefore retain information discarded here: for example multiplier size, changing moduli, or primes outside the old denominator. The square restriction used in the cubic proof concerns precisely a prime at which a middle numerator vanishes.

Theorem 5.7 (sequence form). *Let $a, D, C : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy $D_{n+1} = \text{den}(a_n, D_n)$ and $C_{n+1} = \text{tail}(a_n, D_n, C_n)$. If $E_n = 0$ for all sufficiently large n and $C_{n+1} \neq 0$ for all sufficiently large n , then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .*

Formalised as the [eventual Sylvester recurrence](#). The proof is routine: take the larger of the two thresholds and apply Theorem 5.3 at each later index. Every later section is directed at the hypothesis of this theorem, that is, at forcing E to vanish eventually.

The defect identity has a second consequence, which is what makes a single vanishing error worth having.

Theorem 5.8 (zero is absorbing). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ be an exact natural orbit, so $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$, and let $E_n = \text{ctr}(a_n, D_n, C_n)$. Suppose the centring is strict, $|E_n| < C_n$ for every n . If $E_n = 0$ then $E_{n+1} = 0$.*

Proof. Putting $E_n = 0$ in Theorem 5.1 gives $\Delta_n C_{n+1} = -E_{n+1}$, so C_{n+1} divides E_{n+1} . A multiple of C_{n+1} of absolute value smaller than C_{n+1} is zero, and $|E_{n+1}| < C_{n+1}$ by hypothesis. $\square$

Formalised as the [absorption of a vanishing centred state](#), with the companion [contra-positive along a tail](#): if zero is absorbing beyond some index and E does not vanish eventually, then E is nowhere zero beyond that index.

Absorption changes the shape of the problem. Either E hits zero once and stays there, or it is never zero. Sections 9 and 10 address the special all-negative case only. The mixed-sign analysis in Sections 12 and 13 instead separates cofinally negative errors from an eventually positive tail.

6 Weighted first crossings of LCM records

This section recovers the weighted record criterion used by the short note. It is an exact criterion for Problem 1.1, stated under the hypotheses of that problem and no others, and it is reached by a first-crossing argument rather than by a hypothesis on the negative part. Only steps setting a new LCM numerator record contribute, a fixed LCM-scale baseline may be subtracted, and the remaining mass may be logarithmically discounted.

Evidence. The proof below is ordinary mathematics. The first-crossing arithmetic has formalised components in `LcmRecordExcess.lean`. In addition, the separate release contains [the full real-weight canonical criterion](#) and its natural-weight version. These release sources contain proof bodies, but they are outside the supplied successful main-repository build; a Comparator configuration alone is not a replay receipt. The finite-orbit data retained with the source are diagnostic examples, not a proof of termination for all rational seeds.

The criterion moves the finiteness obligation from every step to record steps, with a fixed baseline subtracted. The inherited eventual-nonnegative case is Koizumi’s Proposition 1(2); the first-crossing proof also treats mixed signs. Proposition 7.15 concerns a deliberately weaker avoidance model. Its walks do not preserve all the prime-by-prime coprimality and exact denominator feedback of a canonical orbit. Consequently it is not an impossibility theorem for every coprimality-based approach. What remains is to derive a record budget from the exact recurrence, or find another contradiction on a nonterminal orbit.

Remaining obligation. Derive finiteness of the displayed sum for one admissible weight and one fixed B from rational denominator feedback, or contradict a nonterminal orbit by another route. The logarithmic discount weakens what suffices; it does not establish it.

The coordinates below also appear in Bado’s preprint of September 2026: after the shift between indexing from 0 and from 1, his least common multiple, his Δ , K , u and g

are the L, M, U, V and ρ defined here, with his denominator parameter equal to q , and his (19) is the update $\rho_n U_{n+1} = U_n - V_n$ [8, Prop. 7.1 and (16)–(20), pp. 6–7].

Set

$$L_n = \text{lcm}(q, a_0, \dots, a_{n-1}), \quad M_n = D_n/L_n, \quad U_n = C_n/M_n, \quad V_n = E_n/M_n.$$

The rational tail has denominator dividing L_n , so U_n and V_n are integers. With $\rho_n = \text{gcd}(L_n, a_n)$, the exact updates are

$$M_{n+1} = M_n \rho_n, \quad \rho_n U_{n+1} = U_n - V_n, \quad V_n = L_n - (a_n - 1)U_n.$$

Write $R_n = \max_{j \leq n} U_j$ and $\mathcal{R} = \{n : U_{n+1} > R_n\}$. Strict centring already gives $U_{n+1} < U_n$ when $\rho_n \geq 2$, so every sufficiently late strict rise has $\rho_n = 1$. The stronger eventual bound $-U_n \leq 2V_n$, supplied by $V_n/U_n = E_n/C_n \rightarrow 0$, gives the quantitative estimate $U_{n+1} \leq 3U_n/4$ when $\rho_n \geq 2$. At a sufficiently late LCM record, $\rho_n = 1$ and the actual jump is $d_n = U_{n+1} - U_n = -V_n > 0$. At a contracting step with $\rho_n \geq 2$ this identity need not hold.

Theorem 6.1 (arithmetic weighted-record dichotomy). *Let a_n, L_n, U_n be positive integers and V_n integers satisfying*

$$\begin{aligned} L_{n+1} &= \text{lcm}(L_n, a_n), & \rho_n &= \text{gcd}(L_n, a_n), \\ \rho_n U_{n+1} &= U_n - V_n, & V_n &= L_n - (a_n - 1)U_n, & -U_n &\leq 2V_n. \end{aligned}$$

Let $f : [1, \infty) \rightarrow [0, \infty)$ be finite and nonincreasing, with $\int_1^\infty f(t) dt = \infty$. For each fixed integer $B \geq 0$,

$$\sup_n U_n < \infty \iff \sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

Proof. A bounded integer running maximum increases only finitely many times, so bounded U_n gives a finite sum. Suppose instead that U_n is unbounded. There are infinitely many record rises, each with $\rho_n = 1$. Its digit is greater than one because $d_n = (a_n - 1)U_n - L_n > 0$. If $r < s$ are record indices, then $a_r \mid L_s$ and $\text{gcd}(a_s, L_s) = 1$, hence $\text{gcd}(a_r, a_s) = 1$. The record digits are therefore distinct and pairwise coprime. For any fixed $B \geq 1$, choose B of them greater than B , and take T after their indices. These old digits $m_0, \dots, m_{B-1}$ all divide L_T .

Put $P = \prod_i m_i$ and choose x by the Chinese remainder theorem with $m_i \mid x + i$. Consider the heights $\tau = x + B + kP$. A first crossing $U_n \leq R_n < \tau \leq U_n + d_n$ is a record step. If $d_n \leq B$, then $U_n \in [\tau - B, \tau)$, so some m_i divides U_n . It also divides L_n , hence divides $d_n = (a_n - 1)U_n - L_n$, contradicting $0 < d_n \leq B < m_i$.

If the step first crosses $h \geq 1$ such heights, their spacing gives $(h - 1)P < d_n$. With $r = d_n - B \geq 1$ and $P \geq B + 1$, we have $d_n = B + r \leq Pr$, whence $h \leq r$. Monotonicity of f now gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \leq (d_n - B)f(U_n).$$

Each height above R_T has one first crossing. Summing, and comparing each interval of length P with its left endpoint, yields the finite bound

$$\sum_{\substack{T \leq n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \geq \frac{1}{P} \int_{R_T+P}^{R_N} f(t) dt. \quad (19)$$

Since $R_n \rightarrow \infty$, the right side diverges for every $B \geq 1$. The case $B = 0$ follows by domination. $\square$

Only the lower centring bound was used. In particular, neither normalised vanishing nor a growth estimate for C_n supplies the CRT moduli: the record digits themselves do so. Normalised vanishing enters in the following application to the original sequence.

Theorem 6.2 (weighted record excess). *Assume the growth and rationality hypotheses of Problem 1.1, and let f be as in Theorem 6.1. The sequence is eventually Sylvester if and only if, for some integer $B \geq 0$,*

$$\sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

Proof. If the endpoint fails, centred zero is never reached on a sufficiently late tail. Integrality gives $1/U_n \leq |V_n|/U_n = |E_n|/C_n \rightarrow 0$, so U_n is unbounded. Theorem 6.1, applied after the centring threshold, forces divergence for every B . Removing a finite prefix changes only finitely many global record terms. Conversely, a Sylvester tail telescopes to $x_n = 1/(a_n - 1)$, so $V_n = 0$ eventually. $\square$

For example, $f(t) = 1/[t \log(et)]$ gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (19) is $P^{-1} \log(\log(eR_N)/\log(e(R_T + P)))$. Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. These are specialisations of one crossing theorem.

The criterion also has an exact expression in the original growth defect. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $\theta_n = E_n/C_n$. The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \quad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Thus the two nonnegative summands $U_n f(U_n)(\gamma_n - B/U_n)_+$ and $(-V_n - B)_+ f(U_n)$ differ by at most $3U_n f(U_n)/a_n$. This is summable because $U_n \leq C_n = \exp(o(n))$, $f(U_n) \leq f(1)$, and $a_n \geq \exp(c2^n)$ eventually for some $c > 0$. Consequently Theorem 6.2 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_n f(U_n) \left(\frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \quad (20)$$

for some B . The original hypotheses do not currently supply this finiteness. In particular, termwise convergence to zero is insufficient. The same equivalence holds directly on the canonical LCM orbit under the original positive, strictly increasing integer sequence, rational reciprocal-sum and near-square-growth hypotheses. Its summand is exactly $U_n f(U_n) \max(a_n^2/a_{n+1} - 1 - B/U_n, 0)$ at record indices and zero elsewhere. For every nonnegative nonincreasing f on $[1, \infty)$ with divergent integral, summability for some $B \in \mathbb{N}$ is equivalent to eventual Sylvester recurrence. The checked comparison uses the sufficient constant 16 in place of the sharper ordinary constant 3 above. It proves $\sum_n C_n/a_n < \infty$ from the canonical tail formula and the ratio test for P_n/a_n^2 , whose consecutive-term ratio tends to zero ($P_n = \prod_{j < n} a_j$). Since $U_n \leq C_n$ and $f(U_n) \leq f(1)$, the comparison error is summable. This transfers the criterion; it does not supply the growth budget from the original hypotheses. The separate-release source [GrowthDebt-Summability.lean](#) contains the factored-growth formulation. It is outside the documented main-repository build. The on-page comparison above is an ordinary proof; no new replay or canonical end-to-end check of this release is asserted.

Also, d_n is the actual jump, including any recovery from a drawdown; it must not be replaced by $R_{n+1} - R_n$ in the crossing proof.

7 Criteria in the record coordinates

The criterion of Section 6 lives in the LCM coordinates $U_n = C_n/M_n$, where old divisors persist. The statements of this section live in the primitive coordinates, where old divisors can be deleted. They are collected here because each is an equivalence, a dichotomy, or an unconditional exclusion, and because each sharpens the profile of a counterexample in a direction the earlier sections do not reach. The underlying tools are the Chinese remainder theorem, persistence of prime-power divisors and weighted first-crossing counting; the LCM compression and the reuse comparison are attributed in Section 6.

Primitive coordinates. Put $G_n = \gcd(C_n, D_n)$, $u_n = C_n/G_n$, $v_n = D_n/G_n$, $\tilde{e}_n = E_n/G_n$ and $h_n = G_{n+1}/G_n$, so that $w_n = a_n u_n - v_n = u_n - \tilde{e}_n$, $h_n u_{n+1} = w_n$, $h_n v_{n+1} = a_n v_n$ and $\gcd(u_n, u_{n+1}) = 1$. The primitive error $\tilde{e}_n$ is signed, and $e_n = -E_n$ of Section 4 remains the magnitude in the unreduced coordinates; neither is Koizumi's rational gap ε_n of Section 3. Write $R_n = \max_{k \leq n} u_k$ and $H_n = \max_{j \leq n} C_j$ for the two running maxima, $s_n = R_{n+1} - R_n$ for the true record increment, $d_n = u_{n+1} - u_n$ at a strict rise, $m_n = (-\tilde{e}_n)_+$, $A_n = R_n m_n / u_n$ for the record-amplified negative error, $\delta_n = (a_n^2/a_{n+1} - 1)_+$ for the growth defect, and $\ell(x) = \log_2 \log_2 \max(4, x)$. The symbol A_n is unrelated to the prefix product A_n of Section 14. A step is *clean* when $h_n = 1$. Call the standing hypotheses of Problem 1.1, read through Koizumi's coordinates together with normalised vanishing, the hypotheses (H), and write (S) for the conclusion that the sequence is eventually Sylvester.

Dictionary with Duverney's reduced parameters. For unit numerators, the eventually reduced parameters in Duverney's Theorem 3.1 and Section 5.2 [2, pp. 285–286, 299–

300] can be taken as

$$p_n = u_n, \quad q_n = w_n = a_n u_n - v_n = h_n u_{n+1}, \quad E_n = G_n(p_n - q_n).$$

Indeed, $\gcd(u_n, w_n) = \gcd(u_n, v_n) = 1$. Reduction of the next tail gives

$$h_n = \gcd(w_n, a_n v_n) = \gcd(w_n, a_n^2), \quad p_{n+1} \mid q_n,$$

and elimination of v_{n+1} gives Duverney's recurrence in these coordinates:

$$a_{n+1} = \frac{p_n}{q_n} a_n^2 - a_n + \frac{q_{n+1}}{p_{n+1}}.$$

Thus $q_n/p_{n+1} = h_n$ is precisely the cancellation factor, and $q_n/p_n \rightarrow 1$. This identifies the reduced parameters, not every possible unreduced choice in the original theorem. A bound on $q_n - p_n$ is a primitive bound; it does not bound $(-E_n)_+$ if G_n grows.

Each statement below carries its own evidence line. A checked statement means a declaration in the successful build covered by the supplied receipt at 3d6d938d696f; a complete proof body in a different release is identified as such, not silently included in that build. A challenge statement containing sorry, a theorem name in a Comparator configuration, and a successful replay are three different kinds of evidence. For an ordinary proof using formalised lemmas, the additional mathematical argument remains part of the proof and its hypotheses are not discharged merely by citing the lemma.

Lemma 7.1 (the denominator valuation transition). *Let u, v, a be positive integers, $\gcd(u, v) = 1$, and $w = au - v > 0$. Put $h = \gcd(w, av)$ and $v' = av/h$. For a prime p , write $r = v_p(a)$, $s = v_p(v)$ and $t = v_p(w)$. Then*

$$v_p(v') = \begin{cases} \max(r, s), & r \neq s, \\ \max(0, 2s - t), & r = s. \end{cases}$$

In particular $v_p(v') \leq \max(r, s)$. A strict loss relative to s requires $r = s \geq 1$ and $t > s$.

Proof. Always $v_p(v') = r + s - \min(t, r + s)$. If $r \neq s$, primitivity implies $t = \min(r, s)$: when $s > 0$, u is a p -adic unit, and when $s = 0 < r$, v is a unit. If $r = s > 0$, both terms in $au - v$ are divisible by p^s , so $t \geq s$; substituting gives the second case. If $r = s = 0$, the same formula gives zero without needing u to be a unit. $\square$

Cancellation is not the same as a fall in the reduced denominator valuation. For example, $(u, v, a) = (2, 15, 9)$ gives $w = 3, h = 3$ and $(u', v') = (1, 45)$. The primitive gap $w - u$ is 1, yet the 3-adic valuation of the denominator rises from 1 to 2. This is a finite exact step, not an infinite counterexample.

Corollary 7.2 (a protected prime power). *Suppose $p^k \mid v_s$. If $w_n < p^{k+1}$ at every step from s through $t - 1$, then $p^k \mid v_t$.*

Proof. At a first loss of divisibility by p^k , the current exponent is some $j \geq k$ and the valuation lemma forces $v_p(w_n) > j$. This would give $w_n \geq p^{j+1} \geq p^{k+1}$, contrary to the hypothesis. $\square$

These are ordinary local calculations. They explain the persistence threshold used below without identifying any new Lean declaration.

7.1 The record-increment coefficient

Theorem 7.3 (record-excess dichotomy). *Let $\Theta = \limsup_n (H_{n+1} - H_n)/\ell(H_n)$ on a non-terminating canonical orbit under (H) . Then either G_n is unbounded and Θ is infinite, or G_n stabilises at a value g and $\Theta \geq g v_T/\varphi(v_T)$ for every late T , so that $\Theta > g \geq 1$. Hence $\Theta = 0$ or $\Theta > 1$, and $\Theta \leq 1$ forces (S) .*

Corollary 7.4 (inclusive log-log boundary). *If $\limsup_n (-E_n)_+/\ell(C_n) \leq 1$ then (S) holds. Every counterexample therefore satisfies $\limsup_n (-E_n)_+/\ell(C_n) > 1$ strictly.*

Mechanism. Four steps. The drawdown ladder $r_n = (-E_n - B_n)_+$ with $B_{n+1} = (B_n + E_n)_+$ and $B_n = H_n - C_n$ gives $0 \leq r_n \leq (-E_n)_+$, so a bound on the record increment is weaker than a bound on the negative error and is therefore the better trigger. A divisor of C_t and D_t at a record locks into every later record increment, so a pairwise-coprime block of moduli above B dividing D_T contradicts $r_n \leq B$ below height $2P + B$. If $\sup_n G_n$ is infinite then Θ is infinite, so a finite Θ forces a genuinely primitive tail on which all multipliers are pairwise coprime. Pre-sieving the CRT block with $x \equiv 0 \pmod{v_N}$ makes a block of length B cost only $\varphi(v_N)v_N^{-1}B$ fresh moduli, which gives the strict inequality.

Evidence. Ordinary proof over the Lean core. The kernel-checked ingredients are the [persistence of a common divisor](#), the [bounded CRT block](#), the [slow-rise landing lemma](#), and the [coprime-block exclusion](#). The running-maximum persistence clause, the record-increment trigger, the gcd rescaling and the pre-sieved amplification are ordinary mathematics. Finite replay: the identities hold exactly on four orbits, the $B = 20$, $R = 6$ CRT reconstruction equals 521026757646 and excludes all twenty positions, and the Fermat product identity and the maximal gap ratio 1.8116 at $u \geq 1000$ against $\sigma^{-1} = 2$ hold.

No upper bound on Θ is proved. If G_n is unbounded, the theorem forces $\Theta = +\infty$. If $G_n = g$ eventually, it proves only $\Theta \geq g v_T/\varphi(v_T)$ for every late T . Writing $\sigma = \lim_T \varphi(v_T)/v_T$, the case $\sigma = 0$ again forces infinity, whereas $\sigma > 0$ leaves a possible finite coefficient at least g/σ . Both that finite-coefficient case and the infinite-coefficient case remain unexcluded. Replacing H_n by the running maximum of u_n divides the coefficient by g , since $\ell(gx)/\ell(x) \rightarrow 1$.

7.2 Two exact equivalences under arbitrary cancellation

Theorem 7.5 (record-amplified rigidity). *Under (H) , the following are equivalent: (S) ; $\limsup_n A_n < \infty$; $\limsup_n R_n \delta_n < \infty$. Each of those two limits superior is 0 or $+\infty$.*

Corollary 7.6 (the critical rate). *Under (H) and $\delta_n = O(1/n)$, with no convergence of $n\delta_n$ assumed, (S) is equivalent to $u_n = O(n)$ and to $(-\tilde{e}_n)_+ = O(1)$. A counterexample at the critical rate therefore has $\limsup_n u_n/n = \infty$ and $\limsup_n (-\tilde{e}_n)_+ = \infty$.*

Mechanism. On a non-Sylvester orbit, arbitrarily late indices carry B distinct primes above a chosen bound dividing v_T , because paid steps have density zero and a block of consecutive clean steps has pairwise-coprime multipliers all dividing the final denominator. A finite alternative between a CRT crossing and a deletion then forces some A_n above any prescribed level. The passage to the growth defect uses $|\delta_n - m_n/u_n| \leq 3/a_n$ and $R_n/a_n \rightarrow 0$.

Ordinary proof. The release contains the cancellation-visibility lemma [record-amplified error after cancellation](#): at the first later negative error, $u_t \leq u_{s+1}$ and $u_s u_t \leq R_t |\tilde{e}_t| u_{s+1}$. This revision also imported the same module into the public `ErdosProblems` root; `AxiomAudit` prints `recordAmplified_error_after_cancellation` against the standard axioms. The Palomar Challenge copy remains the Comparator statement. The exact fixture $15/134 = 1/10 + 1/85 + 1/5695$, with primitive steps $(15, 134) \rightarrow (4, 335) \rightarrow (1, 5695)$ and $A_1 = 15/4$, meets the visibility bound with equality.

Remaining obligation. The equivalence relocates the problem into the A_n axis. Nothing here bounds A_n on the canonical orbit.

Lemma 7.7 (canonical odd-prime-power supply). *Under (H), for every fixed $A > 0$ and every sufficiently large n , the primitive denominator v_n has an odd prime-power divisor $Q = p^k$ with*

$$Q > (H_n + 2)^A, \quad H_n = \max_{j \leq n} C_j.$$

The prime p may depend on n ; no stable-gcd or prime-arrival assumption is imposed.

Proof. The canonical tail estimate gives $C_{n+1}/C_n \rightarrow 1$, hence $\log H_n = o(n)$. Since $x_n = u_n/v_n \leq 2/a_n$ and $u_n \geq 1$, $v_n \geq a_n/2$. Also $v_n \mid L_n = \text{lcm}(q, a_1, \dots, a_{n-1})$, so the 2-primary part of v_n is at most $\max(q, a_{n-1}) = a_{n-1}$ for all large n . Its odd part W_n therefore satisfies

$$W_n \geq \frac{a_n}{2a_{n-1}} \geq \frac{a_{n-1}}{4}, \quad \log W_n \geq c2^{n-1} - O(1)$$

for some $c > 0$. Put $B_n = (H_n + 2)^A = \exp(o(n))$. If every exact odd prime-power factor of W_n were at most B_n , then $W_n \mid \text{lcm}(1, \dots, \lfloor B_n \rfloor)$, giving

$$\log W_n \leq B_n \log B_n = \exp(o(n)).$$

This contradicts the preceding exponential lower bound for all sufficiently large n . An offending exact prime-power factor is the required Q . $\square$

The argument compares the logarithm of the odd denominator part with the logarithm of a factorial bound. It needs neither the prime number theorem nor a lower density of new primes, and makes no such claim.

Theorem 7.8 (unit record increments). *Under (H), if $R_{n+1} - R_n \leq 1$ for all large n , then $a_{n+1} = a_n^2 - a_n + 1$ for all large n . Hence (S) holds if and only if $\#\{n : R_{n+1} - R_n \geq 2\}$ is finite. No hypothesis is placed on drawdowns, on record-setting jumps, or on the cancellation factors h_n .*

Proof. Suppose the orbit is not eventually Sylvester. Absorption gives $\tilde{e}_n \neq 0$ on a late tail, so normalised vanishing and integrality give $u_n \rightarrow \infty$. Choose s beyond the unit-increment and centring thresholds. Lemma 7.7 supplies an odd $Q = p^k \mid v_s$ with $Q > 4(R_s + 2)$ and $Q \geq 16$. Let Y be the least multiple of p above R_s . Then $Y \leq R_s + p < Q/4 + p \leq 5Q/4$. At the first $t > s$ with $u_t \geq Y$, the integral unit-record increment forces $u_t = Y$. Before t , one has $w_n < 3u_n/2 < 15Q/8 < pQ = p^{k+1}$. Corollary 7.2 therefore gives $p^k \mid v_t$. But $p \mid u_t = Y$, contradicting $\gcd(u_t, v_t) = 1$. The converse follows because a Sylvester tail has $u_n = 1$ eventually. $\square$

The separate release contains the [unit-increment landing lemma](#) and the [one-unit record-increment implication](#). This revision imported the same module into the public ErdosProblems root; AxiomAudit prints recordIncrementOne_sylvesterNext_eventually against the standard axioms. Their prime-power supply hypothesis is justified here by the preceding ordinary lemma, not by the inherited receipt; the assembled canonical paper theorem remains ordinary.

Relation. This theorem is incomparable with the two-unit record theorem: a record-setting jump of size at most two bounds $s_n \leq 2$ rather than $s_n \leq 1$, and $s_n \leq 1$ allows arbitrarily large record-setting jumps after a drawdown. The fixture $(8, 177) \rightarrow (7, 4071) \rightarrow (10, 2373393)$ with multipliers 23 and 583 has records 8 and 10, increment 2, jump 3 and $\gcd(8, 10) = 2$, which is why the parity step of the odd cut does not transfer from jumps to increments.

7.3 Protected epochs and the energy criterion

Theorem 7.9 (protected epoch inequality). *Let the orbit satisfy the primitive cocycle with $2|\tilde{e}_n| < u_n$ from an index s . Let $p \geq 3$ be prime, $Q = p^k$ divide v_s with $Q \geq 16$, put $L = pQ/2$, and assume $R_s < L/2$. Assume that $u_t \geq L$ for some $t > s$, and let τ be the first such index, let J be the set of steps in $[s, \tau)$ that first cross at least one odd multiple of p in $(L/2, L]$, and put $X = \sum_{n \in J} (d_n - 2)$. Then every $n \in J$ is a clean record step with $d_n \geq 3$, and $pQ \leq (8p + 8)|J| + 4X + 8p$.*

Proof of Theorem 7.9. For $s \leq n < \tau$, one has $u_n < L$ and $w_n < 3u_n/2 < 3pQ/4 < p^{k+1}$, where $Q = p^k$. Corollary 7.2 gives $Q \mid v_t$ throughout $[s, \tau]$, so $p \nmid u_t$ there. A first crossing of a level above R_s is a record step. If $h_n \geq 2$ then $u_{n+1} = w_n/h_n < 3u_n/4$, so every such step is clean. An odd multiple of p cannot be landed on. A jump of size 1 crossing it would land on it; a jump of size 2 avoiding the landing would have two even endpoints, contrary to $\gcd(u_n, u_{n+1}) = 1$. Thus every $n \in J$ has $d_n \geq 3$. There are at least $Q/8 - 1$ odd multiples of p in $(L/2, L]$. The levels first crossed by a step of size d_n have spacing $2p$, so there are at most $1 + d_n/(2p)$ of them. Summing over J gives

$$Q/8 - 1 \leq |J| + \frac{2|J| + X}{2p},$$

which rearranges to the asserted inequality. $\square$

Theorem 7.10 (energy criterion). *Under (H), the energy $\mathcal{E} = \sum_{n \text{ record}} (\mathbf{1}_{d_n \geq 3} u_n^{-1/2} + (d_n - 2)_+ u_n^{-1})$ is finite if and only if (S) holds; and $\sum_{n \text{ record}} (d_n - 2)_+ u_n^{-1/2}$ is finite if and only if (S) holds.*

Proof. Assume first that the orbit is nonterminal. Then $u_n \rightarrow \infty$. For every sufficiently late s , Lemma 7.7 gives an odd $Q = p^k \mid v_s$ with $Q \geq 16$ and $pQ > 4R_s$. Set $L = pQ/2$ and take its first crossing. The preceding theorem applies. Because $Q \geq p$ and $p \geq 3$,

$$\frac{8p}{L} = \frac{16}{Q} \leq 1, \quad \frac{8p + 8}{\sqrt{L}} \leq 8\sqrt{2}(1 + 1/p) < 16.$$

Dividing the epoch inequality by L therefore gives

$$1 \leq 16 \frac{|J|}{\sqrt{L}} + 4 \frac{X}{L} \leq 16 \sum_{n \in J} \left(\frac{1}{\sqrt{u_n}} + \frac{d_n - 2}{u_n} \right),$$

since $u_n < L$ for $n < \tau$ and $d_n \geq 3$ on J . Thus an arbitrarily late finite window contributes at least $1/16$ to $\mathcal{E}$, contradicting convergence. No disjointness of the windows is needed: every window lies in a tail of the nonnegative series. On a Sylvester tail $u_n = 1$ eventually, so there are no late records and $\mathcal{E}$ is finite. Finally, term by term at a record,

$$\mathbf{1}_{d_n \geq 3} u_n^{-1/2} + (d_n - 2)_+ u_n^{-1} \leq 2(d_n - 2)_+ u_n^{-1/2}.$$

Hence finiteness of the second series implies finiteness of the first, and its converse end-point again follows from eventual Sylvester behaviour. $\square$

Evidence. The separate release contains [the integer protected-epoch inequality](#) and its counting lemmas. This revision imported that module into the public ErdosProblems root; AxiomAudit prints protected_epoch_energy_integer against the standard axioms. The canonical supply, real-valued bound and assembled energy equivalence above remain ordinary paper arguments; the Lean inequality is the checked integer form, not a checked transcription of the whole assembled theorem. The checked main-repository declaration [the two-unit record-rise implication](#) retains a supply premise: arbitrarily late s admit an odd $p^k \mid v_s$ with $3R_s < p^k$. Lemma 7.7 supplies this premise in ordinary mathematics. The declaration itself has not thereby become an end-to-end checked canonical theorem. This distinction is about formal coverage, not an unfilled mathematical supply argument in the text.

7.4 A slow negative part, and the original coordinates

Theorem 7.11 (slow negative part). *Let (a, C, D) be an exact natural orbit with $a_n > 1$, $C_n > 0$, $D_0 \geq 1$, under normalised vanishing. Suppose that for some $\delta \in (0, 1)$ and all large n with $E_n < 0$ one has $-E_n \leq (1 - \delta)\ell(C_n)$. Then $E_n = 0$ for all large n , and $a_{n+1} = a_n^2 - a_n + 1$ for all large n .*

Theorem 12.1 is the case of a constant bound, since a constant is eventually below $(1 - \delta)\ell(C_n)$.

Mechanism. Normalised vanishing gives $C_{n+1} < \frac{3}{2}C_n$, hence $\ell(C_n) \leq \log_2 n + O(1)$, and a_n grows at least geometrically. The multiplier overlap $\gcd(a_n, D_n)$ is bounded by the next negative magnitude and therefore by $\ell(C_n) + 1$. Splitting a_n into the part coprime to D_n and the burnt part, a poor index supplies a burn prime with valuation above $\sqrt{n}$, and such primes are injective across a window, so a window $[T, 2^T]$ has at most $T + O(1)$ poor indices. The first B rich indices give B pairwise-coprime moduli above any prescribed size, and the block form of the landing lemma applies.

Evidence. Ordinary proof over the Lean core. The kernel-checked ingredients are the [persistence of a common divisor](#), the [overlap transport](#), the [landing lemma](#) and the [coprime-block exclusion](#). The burn count and the choice of constants are ordinary mathematics. Corollary 7.4 supersedes this theorem on two axes, the drawdown-corrected

observable and the inclusive constant, and this theorem is retained because its hypothesis is stated directly on E_n .

Theorem 7.12 (product-weighted defect: bounded and slow-growth forms). *Let $a_{n+1}/a_n^2 \rightarrow 1$ and $\sum 1/a_n \in \mathbb{Q}$, and put*

$$Q_n = \frac{a_1 a_2 \cdots a_{n-1}}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right).$$

If $\limsup_n Q_n < \infty$ then the sequence is eventually Sylvester. If for some $\delta > 0$ and all large n one has $Q_n \leq (1 - \delta)\lambda \ell(a_1 \cdots a_{n-1}/a_n)$, where $\lambda > 0$ depends only on the sequence, the same conclusion holds.

For the bounded clause, choose $q > 0$ clearing the reciprocal sum, put $P_n = \prod_{j < n} a_j$, $x_n = \sum_{k \geq n} 1/a_k$, and use the canonical state $D_n = qP_n$, $C_n = D_n x_n$, $E_n = D_n - (a_n - 1)C_n$. With $\gamma_n = a_n^2/a_{n+1} - 1$, the two-term tail estimate gives

$$\begin{aligned} E_n &= \frac{qP_n}{a_n} - \frac{qP_n(a_n - 1)}{a_{n+1}} + O\left(\frac{qP_n a_n}{a_{n+1}^2}\right), \\ q\frac{P_n}{a_n}\gamma_n &= \frac{qP_n a_n}{a_{n+1}} - \frac{qP_n}{a_n}. \end{aligned}$$

The leading terms cancel when these identities are added. Since $P_n/a_n = \exp(o(n))$ and quadratic growth makes a_n doubly exponential,

$$E_n + qQ_n = \frac{qP_n}{a_{n+1}} + O\left(\frac{qP_n a_n}{a_{n+1}^2}\right) = o(1).$$

Thus the extra assumption $\limsup Q_n < \infty$ supplies the eventual lower bound on E_n required by Theorem 12.1. This canonical-tail transfer is given by ordinary analysis here. The bounded product-defect clause also has the checked original-coordinate theorem in `PaperCompleteR7/ProductDefect.lean`; the slow-negative clause requires its separately stated ordinary growth argument.

Attribution. Erdős and Straus require $\limsup \leq 0$ in the corresponding criterion, with the least common multiple in place of the product and the growth factor one index later, so their quantity is $[a_1, \dots, a_n]a_{n+1}^{-1}(a_{n+1}^2/a_{n+2} - 1)$ [1, Theorem 3, p. 132]; Koizumi's Corollary 4(1) uses the product form [12, Cor. 4(1), pp. 14–15]. The bounded clause is Theorem 12.1 transported into that language, and its content over the published criteria is the passage from ≤ 0 to $< \infty$. It is the only place in this note where the machine-checked theorem is written in the vocabulary of the 1964 paper.

Evidence. Ordinary proof. The bounded clause transports Theorem 12.1, and the slow-growth clause also uses Theorem 7.11; both go through the canonical identity $E_n + qQ_n = o(1)$ established above, whose classical predecessor is the comparison in Koizumi's proof of Corollary 4(1) [12, (11)–(12), p. 15].

Corollary 7.13 (the $1/n$ threshold). *Let $a_{n+1}/a_n^2 \rightarrow 1$ and $\sum 1/a_n \in \mathbb{Q}$. If $\limsup_n n(a_n^2/a_{n+1} - 1)_+ < 1$, then $a_{n+1} = a_n^2 - a_n + 1$ for all large n . The same conclusion holds if, for some $K \geq 0$ and $\varepsilon > 0$,*

$$\frac{a_n^2}{a_{n+1}} - 1 \leq \frac{1}{n} + \frac{K}{n^{1+\varepsilon}} \quad \text{eventually.}$$

In particular the one-sided bound by $1/n$ is included.

Proof. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $t_n = (\prod_{j \leq n} a_j)/a_n$, so that $t_{n+1}/t_n = 1 + \gamma_n$. Choose $r \in (0, 1)$ and N with $\gamma_n^+ \leq r/n$ for all $n \geq N$. Then

$$t_n \leq t_N \prod_{k=N}^{n-1} \left(1 + \frac{r}{k}\right) = O(n^r),$$

so $(t_n \gamma_n)_+ = t_n \gamma_n^+ = O(n^{r-1})$ tends to zero and $\limsup_n t_n \gamma_n \leq 0$. Koizumi's Corollary 4(1) [12, pp. 14–15] gives the first conclusion. For the second, positivity of $1 + \gamma_n$ and

$$1 + \gamma_n \leq (1 + 1/n)(1 + K/n^{1+\varepsilon})$$

give $t_n = O(n)$, since the second product converges. Consequently $t_n(\gamma_n)_+ = O(1)$, and the bounded product-defect criterion applies. $\square$

The endpoint is one-sided and quantitative. The weaker statement $\limsup n(\gamma_n)_+ \leq 1$ does not by itself give this product estimate: the scalar template $\gamma_n = (1 + 1/\log n)/n$ has $t_n \asymp n \log n$ and $t_n \gamma_n \asymp \log n$. This is a limitation of the estimate, not a canonical counterexample. The strict clause includes Koizumi's sufficient rate $1 + o(1/n)$ [12, Remark 3, p. 16]; the inclusive clause uses the bounded-defect extension rather than the classical nonpositive upper-limit criterion.

7.5 Pointwise signs and asymptotic criteria

Proposition 7.14 (the classical quantities carry a shifted sign). *The overlap debt divides the tail gcd, so $\kappa'_n = G_n/M_n$ is a positive integer and $E_n/M_n = \kappa'_n \tilde{e}_n$ is an integer carrying the sign of E_n . The growth defect carries a shifted sign. On any exact orbit,*

$$\frac{a_n^2}{a_{n+1}} - 1 = -\frac{E_n}{C_n} + \Lambda_n, \quad \Lambda_n = \frac{(1 - E_n/C_n)(a_n - 1 + E_{n+1}/C_{n+1})}{a_{n+1}}, \quad (21)$$

and $0 < \Lambda_n < 3/a_n$ for all large n whenever $E_n/C_n \rightarrow 0$. The Erdős–Straus quantity of Theorem 3 is

$$Z_n^{\text{ES}} = \frac{[a_1, \dots, a_n]}{a_{n+1}} \left(\frac{a_{n+1}^2}{a_{n+2}} - 1 \right).$$

Its digit LCM includes a_n and does not include the clearing denominator q . It is not $Q_n = (P_n/a_n)\gamma_n$ from the preceding subsection, nor $L_n \gamma_{n+1}/a_{n+1}$ under our convention $L_n = \text{lcm}(q, a_1, \dots, a_{n-1})$. Being a positive multiple of the next growth defect, Z_n^{ES} has the sign of $\Lambda_{n+1} - E_{n+1}/C_{n+1}$. It is positive when $E_{n+1} \leq 0$, and for $E_{n+1} > 0$ it is negative precisely when $E_{n+1}/C_{n+1} > \Lambda_{n+1}$. On a Sylvester tail $E_n = 0$ and $\Lambda_n = (a_n - 1)/a_{n+1} > 0$.

Proof. The first sentence is the divisibility dictionary of Section 7. For (21), write $\theta_n = E_n/C_n$; Proposition 4.1 gives $C_{n+1} = C_n(1 - \theta_n)$ and $D_n/C_n = a_n - 1 + \theta_n$, so $a_{n+1} - 1 + \theta_{n+1} = a_n D_n/C_{n+1} = a_n(a_n - 1 + \theta_n)/(1 - \theta_n)$. Multiplying the asserted identity by a_{n+1} and substituting reduces it to $a_{n+1} + a_n - 1 + \theta_{n+1} = a_n^2/(1 - \theta_n)$, which is the displayed relation with a_n added to both sides. The bound follows from $\theta_n \rightarrow 0$ and $a_{n+1} \geq a_n^2/2$. On Sylvester's sequence $\theta_n = 0$ and $a_{n+1} = a_n^2 - a_n + 1$, so $a_n^2/a_{n+1} - 1 = (a_n - 1)/a_{n+1} = \Lambda_n > 0$. $\square$

Reading. A nonpositive upper limit admits positive values tending to zero; it is not an eventual pointwise sign condition. For the *product* quantity $Q_n = (P_n/a_n)\gamma_n$, the identity $E_n + qQ_n = o(1)$ shows that eventual $E_n \geq 0$ implies $\limsup Q_n \leq 0$, as in Koizumi's Corollary 4(1) [12, pp. 14–15]. The differently indexed LCM quantity Z_n^{ES} is governed by the published hypotheses of Erdős–Straus Theorem 3 [1, p. 132]; the identity with qQ_n must not be reused for it. The earlier 4,472-frame and 610-triple replay counts are inherited finite evidence, not a new check or a proof of an asymptotic criterion.

Proposition 7.15 (what whole-modulus avoidance permits). *Let $m_0 < m_1 < \dots$ be pairwise coprime integers at least 2 with $\theta = \sum_i 1/m_i < 1$. For all integers $x \geq 1$ and $L \geq 1$ satisfying $L > k/(1 - \theta)$, where $k = \#\{i : m_i \leq x + L\}$, the window $[x, x + L)$ contains an integer divisible by no m_i . Consequently, if $\ell(m_i) = i + O(1)$ then for every $\epsilon > 0$ some subfamily carries a walk $u_n \rightarrow \infty$ divisible by no m_i with $u_{n+1} - u_n \leq (1 + \epsilon)\ell(u_n)$ for all large n .*

Proof. The interval $[x, x + L)$ contains exactly L integers. Each lies in $[1, \infty)$ and is smaller than $x + L$, so a modulus exceeding $x + L$ divides none of them. Each of the k remaining moduli divides at most $L/m_i + 1$ of them, so the covered count is at most $L\theta + k < L$.

For the walk, discard an initial segment of the family and reindex so that $\theta < \epsilon/(1 + \epsilon)$. If $k(z) = \#\{i : m_i \leq z\}$, the hypothesis $\ell(m_i) = i + O(1)$ gives $k(z) \leq \ell(z) + C$ for all large z and some constant C . Choose

$$(1 - \theta)^{-1} < \rho < 1 + \epsilon, \quad L(y) = \lceil \rho \ell(y) \rceil.$$

Then $L(y) = O(\ell(y)) = o(y)$, and the definition of ℓ gives $\ell(y + 1 + L(y)) = \ell(y) + o(1)$. Consequently, for all large integers y ,

$$\frac{k(y + 1 + L(y))}{1 - \theta} \leq \frac{\ell(y) + C + o(1)}{1 - \theta} < \rho \ell(y) \leq L(y),$$

while $L(y) \leq (1 + \epsilon)\ell(y)$. Starting with u_0 sufficiently large, apply the first part with $x = u_n + 1$ and length $L(u_n)$, and choose u_{n+1} in the resulting window. The sequence is strictly increasing, avoids every retained modulus, and has the required rise bound. $\square$

The integer hypotheses are used in the first sentence of the proof. For example, the real interval $[99.1, 100.2)$ has length 1.1 but contains only the integer 100; a covered count smaller than 1.1 need not leave an uncovered integer.

Reading. Proposition 7.15 concerns avoidance of whole multiples, not coprimality to a composite modulus. The distinction matters: no integer in $[2, 5)$ is coprime to 30, although none is divisible by 30. Thus this proposition alone does not disprove a coprime-walk extension of Theorem 11.3. Proposition 14.6 below gives a separate counterexample using sparse *prime* moduli. Under the additional scale $\ell(m_j) = j + O(1)$, Theorem 14.7 determines the exact maximal-gap coefficient for whole-modulus avoidance. Its proof replaces the former numerical conjecture; the previously recorded finite value 1.8116 for the Fermat family is not used. Neither static construction supplies an exact reciprocal-tail orbit.

The exact remaining obstacle. Every criterion of this section leaves the same regime, which is

$$(1 - \delta)\ell(C_n) \leq -E_n = o(C_n) \quad \text{at infinitely many } n,$$

with divergent normalised negative mass, unbounded record-amplified error, and infinitely many clean primitive record jumps of size at least three. These criteria do not exclude that regime. The residue-saturation lemma shows precisely one limitation of the fixed-old-modulus method; it is not an impossibility theorem about every future approach. A useful next target is interaction between new prime support, three consecutive numerators and the global height budget.

8 Descent when the error is nonnegative

The hypothesis of the theorem below is the update law of Proposition 4.1 read inside $\mathbb{N}$: the error is nonnegative at every index, so the tail state never rises. This is the easy case, and it is settled in three lines.

Theorem 8.1 (descent). *Let $C, E : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $C_{n+1} + E_n = C_n$ for every n . Then $E_n = 0$ for all sufficiently large n .*

Proof. A one-line descent. The relation forces $C_{n+1} \leq C_n$, so C is a nonincreasing sequence of natural numbers and is eventually equal to its minimum; beyond that index $C_{n+1} = C_n$, and the relation gives $E_n = 0$. $\square$

Formalised as the [stabilisation of the nonnegative error](#), on the [eventual constancy of a nonincreasing sequence](#).

The hypothesis is exactly one-sidedness: C and E take values in $\mathbb{N}$. If $E_n < 0$ at infinitely many indices then C rises there, nothing descends, and the argument is unavailable. That is the whole difficulty, and the rest of this note is directed at it.

9 Excluding a constant negative magnitude

The exclusions in this section and the next concern infinite constant or periodic negative magnitudes, and they do not bound the length of a finite transient. Arbitrarily long constant transients occur on genuine rational orbits, by the flat-transient theorem of the working report on Problem #243 [9, “Flat-transient theorem”].

Suppose the error is negative with a constant magnitude, $E_n = -m$ for a fixed $m > 0$ and every n . By Proposition 4.1 the state then climbs by exactly m each step, so $C_n = c + nm$ with $c = C_0$, and the definition of the centred state becomes a *shape equation*

$$D_n + m = (a_n - 1)(c + nm). \quad (5.1)$$

Together with $D_{n+1} = a_n D_n$ this is a closed system in (a, D) , and it has no solutions. The following instance shows how the failure happens; the theorem then shows that it cannot be avoided.

Example 9.1 (the shape equation running until it fails). Take $m = c = 1$, so that $C_n = 1 + n$ and the shape equation (5.1) reads $D_n + 1 = (a_n - 1)(1 + n)$, and take $D_0 = 1$. Each step is now determined: at $n = 0$ the equation reads $2 = (a_0 - 1) \cdot 1$, so $a_0 = 3$, and $D_1 = a_0 D_0 = 3$; at $n = 1$ it reads $4 = (a_1 - 1) \cdot 2$, so $a_1 = 3$, and $D_2 = a_1 D_1 = 9$; at $n = 2$ it reads $10 = (a_2 - 1) \cdot 3$, which has no integer solution, and the orbit stops. Longer prefixes occur for other starting values: the finite search recorded in Appendix B found none longer than 17 among the initial states below 5000.

Theorem 9.2 (no constant negative magnitude). *There is no pair of sequences $a, D : \mathbb{N} \rightarrow \mathbb{N}$ with $a_n \geq 2$ for all n satisfying $D_{n+1} = a_n D_n$ and (5.1), for any $m > 0$ and any c . The same holds if the shape only begins at some index.*

Proof when c and m are coprime. Assume first $\gcd(c, m) = 1$. Two facts about the multipliers do all the work, and the difficulty lies in the fact that they concern the same finite set of primes from opposite directions: the first makes every multiplier meet that set, the second lets each prime of the set meet at most one multiplier.

Every a_j shares a prime with m . Suppose $\gcd(a_j, m) = 1$. Then m is invertible modulo a_j , so some n has $c + nm \equiv 0$, that is $a_j \mid C_n$; and $a_j \mid D_n$ for every $n > j$, since D is multiplicative with a_j among its factors. Choosing such an n beyond j and reading (5.1) modulo a_j gives $a_j \mid m$, contradicting $\gcd(a_j, m) = 1$ and $a_j \geq 2$.

A prime divisor of m occurs in at most one a_j . Suppose $p \mid m$ and $p \mid a_j$. For $n > j$ we have $p \mid D_n$, so (5.1) gives $(a_n - 1)C_n \equiv 0 \pmod{p}$. Now $C_n = c + nm \equiv c$, and $p \nmid c$ because $p \mid m$ and $\gcd(c, m) = 1$; hence $p \mid a_n - 1$, so $p \nmid a_n$. Thus p cannot divide any later multiplier.

The two facts are incompatible. Each of the infinitely many multipliers needs a prime divisor of m , and each prime divisor of m serves at most one multiplier, while m has only finitely many prime divisors. $\square$

The case $\gcd(c, m) = 1$ is the [scale-primitive exclusion](#). The special case $m = c = 1$, worked through in Example 9.1, where m has no prime divisors at all and the first fact is immediately contradictory, is recorded separately as the [normalised exclusion](#).

Removing the scale. For general c , put $g = \gcd(c, m)$. Equation (5.1) shows $g \mid D_n$ for every n , so dividing D , c and m by g leaves a system of the same shape with coprime data, which the previous case excludes. $\square$

Formalised as the [exclusion at every scale](#), and the eventual form, obtained by shifting the orbit to the first constant index, as the [eventual exclusion](#).

Theorem 9.2 excludes a centred state that is eventually constant and negative, at every magnitude and every scale. It says nothing about a negative state whose magnitude changes, and the argument genuinely uses constancy: the finiteness of the set of prime divisors of m is what the pigeonhole uses, and a varying magnitude presents a fresh set at each index.

10 Excluding a periodic negative magnitude

The next case allows the magnitude to vary, provided it repeats. Write $e_n = -E_n > 0$ for the magnitude, as in Section 4, so that $C_{n+1} = C_n + e_n$; suppose e has period h , meaning $e_{n+h} = e_n$ for every n , and that the state gains a fixed drift $M > 0$ over one period, meaning $C_{n+h} = C_n + M$ for every n .

At $h = 1$ this says that the magnitude is constant and that M is its value, which is the situation of Section 9; Theorem 9.2 already excludes it, and does so without the extra hypothesis $e_n < a_n$ imposed below. The new content is $h \geq 2$.

Section 9 could run its pigeonhole on the prime divisors of the single number m . Here the magnitude changes inside a period, and the number available to run it on is the drift M . Three lemmas replace the two facts of Section 9. The first is that multipliers persist: every a_j divides every strictly later denominator state, the [persistence of a multiplier](#). The second is a *lock*: a prime already dividing D and also dividing the current multiplier is forced into the next tail state, the [one-step lock](#), and once present in both D and C it stays in every later D , C and magnitude, the [persistence of a lock](#). The third transports a lock backwards through the period: if a divisor of the drift M occurs in two multipliers, it is a common divisor of C_0 and of every magnitude, the [repeated-divisor transport](#), using that both the period relation and the drift iterate over any number of periods, the [iterated period](#) and the [iterated drift](#).

Theorem 10.1 (no periodic negative magnitude). *Let $a, D, C, e : \mathbb{N} \rightarrow \mathbb{N}$ with $a_n \geq 2$, $e_n > 0$ and $e_n < a_n$ for every n , satisfying*

$$D_{n+1} = a_n D_n, \quad C_{n+1} = C_n + e_n, \quad D_n + e_n = (a_n - 1)C_n,$$

and suppose $e_{n+h} = e_n$ and $C_{n+h} = C_n + M$ for some $h > 0$ and $M > 0$. This is impossible.

Proof. Strong induction on the drift M . The difficulty lies in the fact that a repeated prime divisor of M need not contradict anything directly. It instead forces a common scale on the whole orbit, which can be divided out, and the induction runs on the drift that the division reduces.

Suppose first that no prime divisor of M is a common divisor of C_0 and of every magnitude. Repeated-divisor transport then applies in the contrapositive: a prime divisor of M occurring in two of the multipliers would be exactly such a common divisor, so each prime divisor of M occurs in at most one multiplier. Against this, the argument of Section 9 run phase by phase (along a fixed residue class modulo h the magnitude is constant and C advances by M at each period, so the multipliers of that phase meet an arithmetic progression exactly as in Section 9) makes every multiplier share a prime with M . Since M has only finitely many prime divisors and there are infinitely many multipliers, pigeonhole gives a contradiction.

Otherwise some prime p divides M , divides C_0 , and divides every magnitude. Then $p \mid C_n$ for every n , since $C_{n+1} = C_n + e_n$ and both summands on the right are divisible by p , and the shape equation $D_n + e_n = (a_n - 1)C_n$ then gives $p \mid D_n$ for every n . Divide D , C and e by p . The multipliers are untouched, so $a_n \geq 2$ and $e_n < a_n$ persist and the magnitudes stay positive; the three recurrences are homogeneous in (D, C, e) and so survive; the period is still h ; and the drift becomes $M/p < M$. The inductive hypothesis applies. $\square$

The first of the two cases above is the [phase-primitive exclusion](#), the induction is the [exclusion at every scale](#), and the eventual form is the [eventual exclusion](#).

The bound $e_n < a_n$ is a genuine restriction and not a normalisation. It is the range in which a multiplier cannot divide its own phase magnitude, which is what the lock argument needs. It holds in the intended reading, where e_n is a centred representative and a_n grows quadratically, but it is assumed here and not derived. The drift hypothesis $M > 0$ is also used: a periodic magnitude with zero drift would be identically zero, which is the case already settled by Section 8.

Remark. On the canonical orbit of [12], read through the dictionary of Section 3, the bound $e_n < a_n$ holds eventually. Lemma 4(2) there gives the centring range $-C_n/2 \leq E_n < C_n/2$, so $e_n = |E_n| \leq C_n/2$ at a negative index, and the proof of Lemma 4(3) gives $C_{n+1} \leq \frac{3}{2}C_n$ [12, pp. 11–12], hence the uniform bound $C_n \leq C_0(3/2)^n$. Against this, summability of $\sum 1/a_n$ forces $a_n \rightarrow \infty$, and the growth hypothesis $a_n^2/a_{n+1} \rightarrow 1$ gives $a_{n+1} \geq a_n^2/2$ for all large n , hence $a_{n+1} \geq 2a_n$ once $a_n \geq 4$; so the multipliers grow at least geometrically with ratio 2 and outgrow the bound on C_n , making e_n/a_n eventually small. Two caveats. Summability is used for this step, so the estimate is not available from the state recurrence alone; and the conclusion is eventual and not universal, whereas Theorem 10.1 imposes $e_n < a_n$ at every index, so it is the eventual form of the exclusion cited above that applies. Under the periodicity hypothesis of Theorem 10.1 the estimate is in any case unnecessary, since a periodic magnitude is bounded while the multipliers tend to infinity.

11 Bounded rise cannot remain coprime to fresh moduli

Periodicity is still a strong assumption. Removing it needs a different kind of obstruction, and the one used here is a counting argument about where an integer sequence with bounded upward steps must land.

Lemma 11.1 (shifted blocks of consecutive multiples). *Let $m_0, \dots, m_{B-1}$ be pairwise coprime and at least 2. For every bound there is a t beyond it with $m_i \mid t + i$ for each $i < B$.*

Proof. Standard Chinese remaindering: solve $t \equiv -i \pmod{m_i}$ simultaneously, then add multiples of $\prod_i m_i$ to pass the bound. $\square$

Formalised as the [shifted consecutive multiples](#).

Lemma 11.1 matches each modulus m_i to its own multiple $t + i$ inside a window of B consecutive integers. Matchings of a set of integers to distinct multiples in an interval are the subject of Erdős Problem #650, solved by van Doorn, Li and Tang: for every set S

of k positive integers, every open interval of length $2 \max S$ contains distinct multiples of at least $\min(k, \lceil 2\sqrt{k} \rceil)$ elements of S , and this count is optimal [11, Theorem 2.1]. Their extremal construction uses the same Chinese remaindering for moduli that need not be pairwise coprime, with their Claim 3.2 supplying the divisibility of residue differences by greatest common divisors that the generalised theorem requires [11, pp. 4–5]. The first-crossing argument below needs only the pairwise coprime case.

Example 11.2 (a block of three consecutive multiples). Take $B = 3$ and $(m_0, m_1, m_2) = (3, 4, 5)$. The congruences $t \equiv 0 \pmod{3}$, $t \equiv 3 \pmod{4}$ and $t \equiv 3 \pmod{5}$ hold exactly when $t \equiv 3 \pmod{60}$. At $t = 3$ the block is $3, 4, 5$ with $3 \mid 3$, $4 \mid 4$ and $5 \mid 5$; at $t = 63$ it is $63, 64, 65$ with $3 \mid 63$, $4 \mid 64$ and $5 \mid 65$. A sequence of natural numbers that starts at 0, tends to infinity and rises by at most 3 at each step cannot step over the block $\{3, 4, 5\}$: it has a first value at least 3, that value is at most 5, and every member of $\{3, 4, 5\}$ is divisible by one of the three moduli.

Theorem 11.3 (bounded rise cannot remain coprime to fresh moduli). *Let $u : \mathbb{N} \rightarrow \mathbb{N}$ tend to infinity with $u_{n+1} \leq u_n + B$ for a fixed integer $B \geq 1$. Then u cannot remain coprime to infinitely many fresh pairwise coprime moduli: there is no family of pairwise coprime $m_i \geq 2$, one for each index, such that $\gcd(m_i, u_t) = 1$ whenever $i < t$.*

The time threshold in the proof is essential. The rise bound restricts upward jumps but permits arbitrary falls: for example, if $M \geq 1$, then $u_0 = M$, $u_1 = 0$, and $u_n = n - 1$ for $n \geq 1$ satisfies the bound with $B = 1$ and tends to infinity, although the large height M is already attained at time 0.

Proof. Choose $m_0, \dots, m_{B-1}$ and use Lemma 11.1 to find $t > \max(u_0, \dots, u_B)$ with $m_i \mid t + i$ for $0 \leq i < B$. Since $u_n \rightarrow \infty$, there is a first $n > B$ with $u_n \geq t$. Minimality and the rise bound give

$$t \leq u_n \leq u_{n-1} + B < t + B.$$

Hence $u_n = t + i$ for some $0 \leq i < B < n$, so $m_i \mid u_n$. The inequality $i < n$ now permits the avoidance hypothesis, which says $\gcd(m_i, u_n) = 1$; this contradicts $m_i \geq 2$. The argument uses the first crossing of the CRT block, not monotonicity of u . $\square$

Formalised as the **bounded-rise barrier**. The key point is that the two hypotheses pull against each other exactly: divergence forces u to travel arbitrarily far, and bounded rise forbids it from stepping over a window of width B . Uniformity of the bound is what the proof uses, and it is the hypothesis that cannot be relaxed by this argument: a rise bounded only along a subsequence leaves the intervening steps free to jump the window. Without a scale restriction on the moduli, Proposition 14.6 gives an unbounded coprime walk with rises $o(\log \log u_n)$. The same forbidden-block crossing, under a two-sided bound on the error, appears in the proof of Bado's Theorem 5.1 [8, pp. 4–5], posted in September 2026.

The moduli are indexed by the same set as the sequence, and the i th of them is asked to divide no value u_t with $t > i$; nothing at all is asked of it at or before its own index. That asymmetry is what makes the hypothesis available in the application below, where m_i is the i th multiplier and only the later numerators are known to be coprime to it.

The barrier applies to the problem because a *reduced* exact tail already carries pairwise coprime moduli. Call (u, v) a reduced exact tail with multipliers a when $\gcd(u_n, v_n) = 1$,

$$u_{n+1} + v_n = a_n u_n, \quad v_{n+1} = a_n v_n.$$

Here u is the numerator and v the denominator: the two recurrences are those of Section 4, with u in the role of the tail state C and v in that of the denominator state D , and with coprimality imposed at every index.

Proposition 11.4 (reduced tails are pairwise coprime). *In a reduced exact tail, a_n is coprime to v_n ; the multipliers at distinct indices are pairwise coprime; and every earlier multiplier is coprime to every later numerator.*

These are the [step coprimality](#), the [pairwise coprimality](#), and the [whole-modulus avoidance](#). Feeding them to Theorem 11.3 gives the form used later: there is no reduced exact tail whose numerator tends to infinity with a uniformly bounded upward increment, the [reduced bounded-rise exclusion](#), together with its eventual form, the [eventual reduced exclusion](#).

Example 11.5 (Sylvester's sequence as a reduced exact tail). Put $u_n = 1$ and $v_n = D_n = 1, 2, 6, 42, 1806, \dots$ as in Example 4.2, with multipliers $a_n = v_n + 1 = 2, 3, 7, 43, 1807, \dots$. Then $\gcd(u_n, v_n) = 1$, $u_{n+1} + v_n = 1 + v_n = a_n u_n$ and $v_{n+1} = a_n v_n$, so this is a reduced exact tail, and Proposition 11.4 returns the classical fact that Sylvester's numbers are pairwise coprime. Its numerator is constant, so it satisfies every hypothesis of the exclusion just stated except divergence; since the tail exists, that hypothesis cannot be dropped.

An arbitrary orbit of Section 4 need not be reduced, so one more step is needed before the barrier can be applied to it: the common factor must stop changing. We call a set of indices *cofinal* when it is infinite, and say that a property holds *cofinally* when the set of indices at which it holds is cofinal. The step below is the only one connecting the orbit of Section 4 to the reduced tails of Proposition 11.4, and it is where the cofinal boundedness in the proof of Theorem 12.1 is used.

Proposition 11.6 (the tail gcd stabilises). *Let (a, D, C) be an exact natural orbit, that is, a triple of $\mathbb{N}$ -valued sequences with $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$, whose centred state $E_n = \text{ctr}(a_n, D_n, C_n)$ is negative cofinally, with magnitudes bounded along that cofinal set. Then $\gcd(C_n, D_n)$ is eventually constant, and beyond that index the orbit divided by the stable gcd is a reduced exact tail.*

Proof. The tail gcd $\gcd(C_n, D_n)$ divides its successor, the [gcd monotonicity](#), so the tail gcds form a positive divisibility chain. At an index where the centred state is negative the tail gcd is exactly $\gcd(C_n, e_n)$, the [gcd at a negative index](#), and is therefore at most the magnitude there; so the chain is bounded along the cofinal set of negative indices. A positive divisibility chain whose values are bounded along a cofinal set is eventually constant, the [chain stabilisation](#). Hence cofinally bounded negative magnitudes make the tail gcd stabilise, the [gcd stabilisation](#), and beyond that point dividing by the stable gcd leaves the numerator and denominator coprime, that is, a reduced exact tail. $\square$

The stabilisation of the tail gcd is the checked statement cited at the end of the proof; the last clause, that the divided orbit is a reduced exact tail, is exposition and carries no separate label. Boundedness is required only *along a cofinal set*. That is what the situation gives: the tail gcd is controlled only at the negative indices, and cofinally many of them is all the divisibility chain needs.

Normalised vanishing by itself gives a different, strictly weaker conclusion. For an exact natural orbit with $C_n > 0$, put

$$G_n = \gcd(C_n, D_n), \quad \Gamma(N) = \#\{0 \leq j < N : G_j < G_{j+1}\}.$$

Proposition 11.7 (normalised vanishing makes strict gcd changes sparse). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$ with $C_n > 0$, and put $E_n = D_n - (a_n - 1)C_n$, $G_n = \gcd(C_n, D_n)$ and $\Gamma(N) = \#\{0 \leq j < N : G_j < G_{j+1}\}$. If $|E_n|/C_n \rightarrow 0$, then $\Gamma(N) = o(N)$. Moreover, for every starting bound B and block length L , some $n \geq B$ satisfies*

$$G_n = G_{n+1} = \dots = G_{n+L}.$$

The first assertion is the [sublinear strict-growth theorem](#); the second is the [arbitrarily late constant-block theorem](#). The proof first converts normalised vanishing into subexponential growth of C_n , then charges every strict divisibility increase of G_n against a finite power-of-two budget. Sparse strict changes force long finite gaps between them.

The quantifiers matter. Proposition 11.6 uses cofinally bounded negative magnitudes and yields eventual constancy. Proposition 11.7 uses only normalised vanishing and yields arbitrarily late constant blocks of each prescribed finite length. It does not yield one infinite constant tail, and it does not exclude cofinally unbounded negative excursions.

12 Excluding a bounded negative part

Assembling the previous section gives Theorem 12.1, the endpoint of the barrier argument. Theorem 13.1 in the next section uses a different finiteness hypothesis, finite *normalised negative mass*, by which we mean convergence of $\sum_n (-E_n)_+ / C_n$, where $x_+ = \max(x, 0)$. The two hypotheses ask for different quantities to be estimated, and Theorem 13.1 has the broader ambient setting, since it assumes only the state recurrence with $C_n > 0$. On the common class of exact orbits with normalised vanishing each of them is equivalent to eventual vanishing of the error. For a scalar sequence they differ: hypothesis (5) below holds for a constant negative part $-b$, which gives $C_n \sim bn$ and a divergent normalised mass, and such a sequence is excluded as an orbit by Theorem 9.2. The hypotheses of Theorem 12.1 are listed in full to distinguish the abstract arithmetic implication from its canonical application. Normalised vanishing is derived for the canonical orbit and implies eventual strict centring. The bounded negative part is the extra hypothesis that the unrestricted problem does not supply.

Informally, hypotheses (5) and (6) below pull against each other. By Proposition 4.1, hypothesis (5) caps the rise of the tail state at B per step, while hypothesis (6), once the error is nowhere zero, forces the tail state to diverge. Proposition 11.6 makes the orbit reduced, and Theorem 11.3 then converts the tension into a contradiction: an integer sequence tending to infinity with upward steps of size at most B cannot remain coprime to

the pairwise coprime moduli a reduced tail carries. The complementary case, in which the error is eventually positive, is the descent of Theorem 8.1.

Theorem 12.1 (bounded negative part). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ and $E : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy*

1. $a_n > 1$ and $C_n > 0$ for every n ;
2. the exact dynamics $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$;
3. $E_n = \text{ctr}(a_n, D_n, C_n)$ for every n ;
4. eventual strict centring: $|E_n| < C_n$ for all large n ;
5. eventually bounded negative part: $-B \leq E_n$ for all large n , for some integer $B \geq 0$;
6. normalised vanishing: for every integer $K \geq 1$ there is an N with $K|E_n| < C_n$ for all $n \geq N$.

Then $E_n = 0$ for all sufficiently large n .

Proof. Suppose not. Absorption removes every zero after a sufficiently late centring threshold. If negative indices are cofinal, their bounded magnitudes make $G_n = \gcd(C_n, D_n)$ stabilise to an integer $g > 0$. Then $u_n = C_n/g$, $v_n = D_n/g$ form a reduced exact tail. Normalised vanishing and $|E_n| \geq 1$ imply $C_n \rightarrow \infty$, hence $u_n \rightarrow \infty$, whereas $u_{n+1} - u_n = -E_n/g \leq B$. For $B \geq 1$ Theorem 11.3 is a contradiction; for $B = 0$ a non-increasing positive sequence cannot diverge. If negative indices are finite, the eventual nonnegative error gives integer descent and eventual zero, also a contradiction. $\square$

Formalised, with hypotheses (4) and (5) holding from some index as stated, as the [eventual bounded-negative-part rigidity](#). The Lean proof applies the [bounded-negative-part rigidity](#), in which the centring and the bound hold at every index, to the orbit shifted past both thresholds. The two auxiliary results it uses are the [divergence from normalised vanishing](#) and the [exclusion of cofinally bounded negative magnitudes](#), the latter over the [tail-divergence form](#).

Corollary 12.2. *Under the hypotheses of Theorem 12.1, together with $C_{n+1} \neq 0$ for all large n , the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .*

Proof. Theorem 12.1 then Theorem 5.7. $\square$

Hypotheses (1)–(3) are the state system, and (6) is the division-free form of $|E_n|/C_n \rightarrow 0$. Hypothesis (4) is logically redundant: hypothesis (6) with $K = 1$ gives $|E_n| < C_n$ eventually. It is retained because the displayed statement tracks the linked Lean declaration; shifting to the resulting threshold gives the same conclusion without an independent centring input. The state declaration takes normalised vanishing as a hypothesis. The separate checked canonical transfer derives it from the rational reciprocal sum and $a_{n+1}/a_n^2 \rightarrow 1$. Thus Theorem 12.1 is an abstract implication whose analytic premises have a separate formal proof. Combined with Koizumi’s bridge below, however, Corollary 12.2 settles the bounded-negative canonical case of Problem 1.1. What is unconditional in this note is the all-negative constant and periodic exclusions, Theorems 9.2 and 10.1, and the barrier itself, Theorem 11.3.

The conditionality can nevertheless be located precisely, because the missing inputs are supplied elsewhere. With the results of [12], Theorem 12.1 becomes a conditional theorem about Problem 1.1 itself rather than about the state system alone. Let (a_n) satisfy the hypotheses of Problem 1.1. After deleting a finite prefix, Corollary 3 of [12, p. 9] makes the sequence the pseudo-greedy expansion of its own reciprocal sum, and Lemma 4 there supplies the integers c_n, d_n, e_n of Section 3. Hypothesis (1) then holds: c_n is a positive integer by Lemma 4(1), and $a_n > 1$ after a further finite shift, because summability of $\sum 1/a_n$ forces $a_n \rightarrow \infty$. Hypothesis (2) is the pair of recurrences $c_{n+1} = a_n c_n - d_n$ and $d_{n+1} = a_n d_n$ of Lemma 4(2), and hypothesis (3) is that lemma's $a_n = (d_n - e_n)/c_n + 1$ read as $e_n = d_n - (a_n - 1)c_n$, which is the map ctr. The centring range $-c_n/2 \leq e_n < c_n/2$ in the same lemma gives hypothesis (4) at every index, which is stronger than the eventual form used here. Hypothesis (6) is the vanishing of the gap sequence in Corollary 3, since $\varepsilon_n = e_n/c_n$. The sole hypothesis not supplied is (5), the eventual bound on the negative part; nothing in [12] yields it and nothing here does either, so Problem #243 stays open.

13 Excluding finite normalised negative mass

Theorem 12.1 uses a uniform bound on the negative part. A different finiteness hypothesis, on the total normalised negative mass, removes a further family of orbits, and the argument is multiplicative rather than arithmetic: it compares C with a convergent product instead of with a system of congruences.

Theorem 13.1 (finite normalised negative mass). *Let $C_n \in \mathbb{N}_{>0}$ and $E_n \in \mathbb{Z}$ satisfy $C_{n+1} = C_n - E_n$. If*

$$\sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} < \infty,$$

then $E_n = 0$ eventually. For an exact reciprocal-tail orbit this implies $a_{n+1} = a_n^2 - a_n + 1$ eventually. No normalised-vanishing hypothesis is required.

Proof. Put $\delta_n = (-E_n)_+/C_n$. The update gives $C_{n+1} \leq C_n(1 + \delta_n)$, so

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n) \leq C_0 \exp\left(\sum_n \delta_n\right).$$

Choose an integer upper bound K for C_n . Each strict rise of the integer sequence contributes at least $1/K$ to $\sum_n \delta_n$, so there are only finitely many rises. The remaining positive integer sequence is nonincreasing and therefore stabilises, forcing $E_n = 0$. The recurrence follows from Theorem 5.7, since $C_n > 0$. $\square$

The scalar conclusion is [the scalar eventual-vanishing theorem](#); its exact-orbit consequence is [the scalar finite-mass recurrence theorem](#). The separate release contains a [canonical finite-mass formulation](#), outside the main-repository build identified here. Summability remains an assumption; the elementary scalar implication needs neither growth nor normalised vanishing.

For the gap sequence of the pseudo-greedy expansion, the same criterion, with the same product bound and integer descent, appears in the Erdős Problem a Day working report on Problem #243, dated 12 August 2026 [9, “A global termination criterion”]. Bado’s Theorem 11.1 is a compensated criterion that keeps a favourable term from prime-support reuse [8, Thm. 11.1 and Remark 11.3, p. 9].

The transfer of the previous section applies here as well. On the canonical orbit of Corollary 3 of [12, p. 9] the recurrence $C_{n+1} = C_n - E_n$ holds with $C_n > 0$ by Lemma 4 [12, pp. 11–12]. Thus Theorem 13.1 is a conditional theorem about Problem 1.1 whose sole remaining hypothesis is summability of $\sum_n (-E_n)_+ / C_n$. That hypothesis is not supplied there, and is the content of Problem 14.5 below.

Example 13.2 (the product bound at a geometric rate). Suppose $C_0 = 100$ and $(-E_n)_+ / C_n \leq 2^{-n-1}$ for every n , so that the normalised negative mass is at most 1; the constraint at $n = 0$ still permits E_0 as negative as -50 . Since $1 + x \leq e^x$,

$$C_N \leq 100 \prod_{n < N} (1 + 2^{-n-1}) \leq 100e < 272$$

for every N . For all sufficiently large n , we have $2^{-n-1} < 1/272$. A strict rise would then force $(-E_n)_+ / C_n \geq 1/C_n > 1/272$, a contradiction. Thus C_n is eventually nonincreasing; as a positive integer sequence it stabilises, and $E_n = 0$ thereafter. The constant 272 comes from the total mass and not from any single magnitude, which is why the hypothesis of Theorem 13.1 is summability of $(-E_n)_+ / C_n$ and not a bound on it.

14 Complements and further questions

The results proved here do not resolve Problem #243. The remaining conditions are stated below without treating an equivalent criterion as a proof.

Each exclusion holds only in its stated regime. Theorems 9.2 and 10.1 exclude the constant and the stated periodic magnitudes on an all-negative tail, and do not exclude those patterns merely along the negative indices of a mixed-sign tail; Theorem 12.1 excludes a bounded negative part under normalised vanishing, and Theorem 13.1 excludes finite normalised negative mass with no normalised-vanishing hypothesis. Together with absorption and descent they fix the profile of any counterexample.

Proposition 14.1 (frontier profile for a counterexample). *For the canonical integer state attached to any counterexample to Problem 1.1,*

$$E_n \neq 0 \quad \text{eventually,} \quad \frac{|E_n|}{C_n} \rightarrow 0,$$

and

$$\limsup_{\substack{n \rightarrow \infty \\ E_n < 0}} (-E_n) = \infty, \quad \sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} = \infty.$$

In particular, negative indices occur infinitely often. Thus the remaining regime consists of globally coherent, unbounded negative excursions with divergent normalised negative mass.

Proof. Koizumi's canonical-tail transfer gives strict centring and normalised vanishing after a finite shift; the strict-centring hypothesis of Theorem 12.1 also follows from normalised vanishing with $K = 1$. Absorption then makes $E_n \neq 0$ eventually, since eventual zero would give the Sylvester recurrence. Descent excludes an eventually nonnegative state. Theorem 12.1 excludes a bounded negative part, and Theorem 13.1 excludes finite normalised negative mass. These statements are unchanged by deleting a finite prefix. $\square$

Lean checks Proposition 14.1 as [the canonical frontier profile](#), from the rational reciprocal sum, the quadratic growth limit, and the failure of eventual Sylvester behaviour. The divergent normalised negative mass is recorded there as divergence of the partial sums.

Put

$$\delta_n := \frac{(-E_n)_+}{C_n}.$$

Every argument in Sections 9–12 uses a finiteness hypothesis: a fixed set of prime divisors, a fixed period, or a fixed bound on the negative part. Theorem 13.1 uses $\sum_n \delta_n < \infty$. Proposition 14.1 records that a counterexample has none of them; it is a frontier statement, not an additional problem equivalent to the original one. Stated at the level of state orbits, the surviving obstruction is the following.

Problem 14.2 (unbounded divergent-mass negative excursions). Exclude, or construct, an exact natural orbit (a, D, C) with $a_n > 1$ and $C_n > 0$, whose multipliers satisfy $\lim_n a_{n+1}/a_n^2 = 1$ and whose centred state satisfies normalised vanishing, and which is negative infinitely often with magnitudes unbounded along that cofinal set and

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

An excursion of this kind has none of the finiteness hypotheses just listed. By Proposition 14.1 the canonical state of a counterexample is such an orbit after deletion of a finite prefix, so an exclusion would settle Problem #243. A proposed construction must be global, and it must supply the growth condition of Problem 1.1 together with a vanishing tail ratio, under which the [reciprocal-series realisation](#) makes the orbit the canonical orbit of the rational number C_0/D_0 . Finite admissible prefixes alone do not provide such a construction.

Remark. The hypotheses in Problem 14.2 beyond the state recurrence are not decorative: the recurrence alone admits unbounded divergent-mass excursions. Put

$$C_n = 2^n, \quad b_0 = 2, \quad b_{n+1} = \frac{1}{2}b_n(b_n + 2), \quad a_n = b_n + 2, \quad D_n = b_n C_n,$$

so that $(b_n) = 2, 4, 12, 84, 3612, \dots$ and $(a_n) = 4, 6, 14, 86, 3614, \dots$. Every b_n is a positive even integer, since $b_n = 2k$ gives $b_{n+1} = 2k(k + 1)$, so all four sequences are $\mathbb{N}$ -valued. The orbit is exact: $C_{n+1} + D_n = 2^n(2 + b_n) = a_n C_n$ and $D_{n+1} = b_{n+1} 2^{n+1} = b_n(b_n + 2)2^n = a_n D_n$. Its centred state is

$$E_n = D_n - (a_n - 1)C_n = b_n 2^n - (b_n + 1)2^n = -C_n,$$

so the negative magnitudes 2^n are unbounded and $\sum_n (-E_n)_+ / C_n$ diverges. What fails is the centring: $|E_n| = C_n$ at every index, so strict centring fails at the boundary and normalised vanishing fails outright. The multipliers satisfy $a_{n+1} = \frac{1}{2}(a_n^2 - 2a_n + 4)$, so $a_{n+1}/a_n^2 \rightarrow \frac{1}{2}$ and the growth hypothesis fails as well. The example therefore says nothing about Problem 1.1, and it does not isolate the centring hypotheses: $E_n = -C_n$ at every index forces that multiplier recurrence, so growth fails with them. It is recorded only to show that the state recurrence alone is not enough.

14.1 Global height and old-factor overlap

The main arithmetic gap is global. Define the cumulative digit LCM and its overlap quotient by

$$L_0 = D_0, \quad L_{n+1} = \text{lcm}(L_n, a_n), \quad M_n = \frac{D_n}{L_n}.$$

Since $D_n = D_0 \prod_{j < n} a_j$, the quotient is integral and $M_n L_n = D_n$. It records the multiplicity lost when the full product scale is compressed to an LCM.

14.2 Repair entropy across a recovery

The cumulative quotient does not by itself distinguish one large deletion from many small ones. The reduced recurrence does. Let $u, h, c : \mathbb{N} \rightarrow \mathbb{N}$ and $e : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy

$$h_n u_{n+1} = u_n - e_n, \quad u_n > 0, \quad h_n > 0.$$

Fix $K, r, L \in \mathbb{N}$ with $K, L > 0$, assume

$$K|e_{r+i}| < u_{r+i} \quad (0 \leq i < L), \quad u_r \leq u_{r+L},$$

and let R be a finite family of moduli. Suppose every $m_q, q \in R$, divides the deletion product $\prod_{r \leq n < r+L} c_n$, and that $c_n^2 \mid h_n$ throughout the interval. Then

$$K^L \text{lcm}(m_q : q \in R)^2 < (K+1)^L. \quad (22)$$

This is the [repair-entropy inequality](#).

The square in (22) is the key point. Exact denominator reduction charges c_n^2 to the scale growth, so a repaired family whose LCM is at least $2^{|R|}$ satisfies

$$K^L 4^{|R|} < (K+1)^L.$$

The formal consequence is the [independent-repair cardinality bound](#). It rules out positive-density deletion of independent old moduli inside recoveries with a fixed relative-error budget.

There is also an endpoint that does not mention a chosen modulus family. If $K|e_n| < u_n$ eventually for every K , then for each fixed $L > 0$ there is an N such that every recovery $u_r \leq u_{r+L}$ with $r \geq N$ satisfies

$$\prod_{0 \leq i < L} h_{r+i} = 1.$$

Thus a surviving nontrivial reset must have recovery length tending to infinity. Lean checks this as [eventual disappearance of fixed-length reset payments](#). The theorem does

not bound a recovery length that varies with r , nor does it supply a lower bound for the LCM of the moduli repaired in such an interval. Those are the missing inputs needed to turn (22) into a global contradiction.

Problem 14.3 (global overlap-height growth). For every nonterminal canonical orbit satisfying Problem 1.1, must

$$\limsup_{n \rightarrow \infty} \frac{\log M_n}{n} > 0?$$

Equivalently, must there be a $K \geq 1$ for which

$$2^n \leq M_n^K$$

at infinitely many indices?

The two displayed formulations are equivalent up to changing the positive constant; the second is not a weaker target. On the canonical orbit, Section 6 supplies $M_n \mid C_n$ and $M_n \leq C_n$. Therefore a positive answer would contradict the subexponential tail-height budget and prove Problem #243. The missing input is the proposed lower bound on overlap growth.

There is also a more local-looking question whose content is nevertheless the entire prefix. Put $A_n = \prod_{j < n} a_j$, so $D_n = D_0 A_n$. From

$$D_0 A_n = (a_n - 1)C_n + E_n$$

one immediately obtains

$$\gcd(A_n, a_n - 1) \mid E_n.$$

The checked [tail-height estimate](#) and normalised vanishing make $|E_n|$ subexponential in n .

Problem 14.4 (old-factor overlap). In every nonterminal canonical orbit satisfying Problem 1.1, is

$$\limsup_{n \rightarrow \infty} \frac{\log \gcd(A_n, a_n - 1)}{n} > 0?$$

Equivalently, do there exist $\eta > 0$ and infinitely many n such that $\gcd(A_n, a_n - 1) \geq e^{\eta n}$?

A positive answer contradicts the displayed divisibility and the subexponential error budget. Arbitrarily long locally admissible blocks do not answer this question: the gcd uses the complete prefix.

14.3 The direct analytic question

Problem 14.5 (summability from rationality). Let (a_n) satisfy Problem 1.1, and let (D_n, C_n, E_n) be its canonical integer state. Must

$$\sum_{n=0}^{\infty} \delta_n = \sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} < \infty?$$

An affirmative answer closes Problem #243 by Theorem 13.1. Since $\delta_n \rightarrow 0$, convergence is equivalently expressible as boundedness of the partial products $\prod_{n \leq N} (1 + \delta_n)$; this is the same criterion in multiplicative form. A negative answer requires a sequence satisfying the full growth and rationality hypotheses. A locally admissible state orbit is insufficient. By Theorem 6.2, it is enough to establish the finiteness of (20) for one decreasing weight with divergent integral and one fixed baseline. This replaces the all-index mass by a discounted series supported only at LCM records.

Remark. One further question is not left open here, since it is answered in [12]: whether normalised vanishing follows from $a_{n+1}/a_n^2 \rightarrow 1$ and rationality. It does. After deleting a finite prefix, Koizumi's Corollary 3 [12, p. 9] makes the sequence the pseudo-greedy expansion of its own reciprocal sum and gives $\varepsilon_n \rightarrow 0$ under exactly the hypotheses of Problem 1.1; rationality is not needed for that step, only summability of $\sum 1/a_n$. Since $\varepsilon_n = E_n/C_n$ under the intended reading of Section 4, this is normalised vanishing. Strict centring comes from the same source and is stronger than what Theorem 12.1 assumes: for a rational reciprocal sum, the range $-c_n/2 \leq e_n < c_n/2$ of his Lemma 4(2) [12, pp. 11–12] gives $|E_n| < C_n$ at every index, which is stronger than eventual strict centring. The eventual form also follows directly from normalised vanishing with $K = 1$. What remains unsupplied is hypothesis (5), the bound on the negative part, which is why the theorem is still conditional.

14.4 A barrier extension

Proposition 14.6 (no scale-free small-rise extension). *There exist strictly increasing primes p_i and a strictly increasing positive integer sequence $u_n \rightarrow \infty$ such that $\gcd(u_n, p_i) = 1$ for all i, n and*

$$u_{n+1} - u_n = O(\sqrt{\log \log(u_n + e^e)}) = o(\log \log(u_n + 3)).$$

Thus the proposed variable-rise barrier is false without a quantitative restriction on the sizes or arrival of the moduli.

Proof. Choose increasing primes $p_i > \max\{\exp(\exp((i+2)^2)), 2^{i+3}\}$, for $i \geq 0$. Primes of arbitrarily large size suffice; no distribution theorem is used. Then $\theta = \sum_i 1/p_i < 1/4$ and $k(z) = \#\{i : p_i \leq z\} \leq \sqrt{\log \log z}$ whenever z is large. For integer x put $L(x) = \lceil 4\sqrt{\log \log(x + e^e)} + 8 \rceil$. Since $L(x) = o(x)$, eventually $L(x) > k(x + L(x))/(1 - \theta)$. The elementary window bound of Proposition 7.15 leaves an integer in $[x, x + L(x))$ divisible by no p_i . The set of such integers is therefore unbounded. Enumerate it increasingly as (u_n) and apply the same bound with $x = u_n + 1$ to obtain $u_{n+1} - u_n \leq L(u_n + 1)$. For prime moduli nondivisibility is exactly coprimality, proving every claim. The moduli are deliberately much sparser than the scale used in the next theorem. $\square$

Theorem 14.7 (the static maximal-gap coefficient). *Let $m_0 < m_1 < \dots$ be pairwise coprime integers at least 2 with $\ell(m_j) = j + O(1)$, where $\ell(x) = \log_2 \log_2 \max(4, x)$. Let $\sigma = \prod_j (1 - 1/m_j) > 0$ and enumerate the positive integers divisible by no m_j as (u_n) . Then*

$$\limsup_{n \rightarrow \infty} \frac{u_{n+1} - u_n}{\ell(u_n)} = \sigma^{-1}.$$

This is a statement about whole-modulus avoidance. It is not a statement about coprimality to composite m_j , or about canonical state orbits.

Proof. Write $k(z) = \#\{j : m_j \leq z\} = \ell(z) + O(1)$. For a fixed prefix length T , set

$$M_T = \prod_{j < T} m_j, \quad \sigma_T = \prod_{j < T} (1 - 1/m_j), \quad \theta_T = \sum_{j \geq T} 1/m_j.$$

Here $\theta_T \rightarrow 0$. By the Chinese remainder theorem, avoiding the first T whole moduli selects precisely the fraction σ_T of the residues modulo M_T .

For the upper bound, an integer interval $[x, x+L)$ contains at least $\sigma_T L - M_T$ integers avoiding that prefix. The remaining moduli cover at most $L\theta_T + k(x+L)$ integers. Choose T large enough that $\sigma_T > \theta_T$, and then any $c > (\sigma_T - \theta_T)^{-1}$. For $L = \lceil c\ell(x) \rceil$, the number left is positive for all sufficiently large x , since $k(x+L) = \ell(x) + O(1)$. Thus the set being enumerated is unbounded, and every sufficiently late such interval meets it. Applying this at $x = u_n + 1$ gives

$$\limsup_n \frac{u_{n+1} - u_n}{\ell(u_n)} \leq (\sigma_T - \theta_T)^{-1}.$$

Letting $T \rightarrow \infty$ gives the upper bound σ^{-1} .

For the lower bound, fix T and let the integer length L tend to infinity. Among the offsets $0 \leq j < L$, precisely $K_L = \sigma_T L + O(M_T)$ avoid the first T moduli. Assign to these offsets distinct moduli $m_T, \dots, m_{T+K_L-1}$. Solve simultaneously $x \equiv 0 \pmod{M_T}$ and $x \equiv -j$ modulo the modulus assigned to j . Put $Q_L = \prod_{i < T+K_L} m_i$ and choose this solution in $[Q_L, 2Q_L)$. Every integer of $[x, x+L)$ is then divisible by a modulus: either an old one or its assigned new one. Let u be the greatest admissible integer below x ; its successor is at least $x+L$, so the gap is at least L . The scale assumption gives

$$\ell(2Q_L) \leq T + K_L + O(1),$$

because $\log_2 m_i = 2^{i+O(1)}$ and these upper bounds sum geometrically. Since $u < 2Q_L$, the ratio for this gap is at least $L/(T + K_L + O(1))$, tending to σ_T^{-1} . These u tend to infinity: $x \geq Q_L \rightarrow \infty$ and admissible integers are unbounded. Hence the limsup is at least σ_T^{-1} for every T . Let $T \rightarrow \infty$ to finish. $\square$

For the Fermat moduli $m_j = 2^{2^j} + 1$, pairwise coprimality and the finite-product identity

$$\prod_{j=0}^N \left(1 - \frac{1}{2^{2^j} + 1}\right) = \frac{2^{2^{N+1}-1}}{2^{2^{N+1}} - 1}$$

give $\sigma = 1/2$, so the coefficient is exactly 2. The two results above answer the auxiliary questions in the previous draft by ordinary proofs. Neither constructs the positive integer multipliers, integral transports, normalised error or reciprocal-tail realisation required for an example in Problem #243. In particular the static density σ must not be substituted for the canonical prefix quantity $\varphi(v_T)/v_T$ without a separate argument.

Formalisation targets

Erdős–Straus. Formalise Theorem 3 of [1], with its prefix-LCM quotient and correctly indexed growth factor, under the published analytic hypotheses.

Duverney. Formalise Corollary 3.2 of [2], including its signed numerators and its stated convergence hypothesis for the signed defect series, and then recover the all-positive specialisation used here. These are useful verification projects, but neither is an additional open mathematical problem.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. Proof authority remains with those sources; this manuscript provides navigation and exposition.

The supplied receipt records a successful build of its enumerated main-repository modules at the fixed revision. This review has not replayed Lean, audited the transitive axioms of every declaration, or checked the separate release as a whole. A source link is not itself a build receipt. In particular the external formal-conjecture entry is a problem statement, not a proof of Problem #243. The new ordinary proofs, exposition, citation choices and interpretation require mathematical review independently of those build records.

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Acknowledgements. The problem numbering follows the Erdős Problems catalogue [13]. The supplied snapshot is historical; no newly verified live catalogue status is asserted here.

A Guide to the formal sources

The checked main-repository links use revision 3d6d938d696f; release-only links state their separate repository and revision. The principal state theorem is in `Reciprocal-TailRigidity.lean`, the scalar finite-mass theorem in `SparseResetRecovery.lean`, and the entropy inequalities in `RepairEntropy.lean`. The canonical analytic transfers are separate checked declarations in `PaperCompleteR7`; their existence must not be inferred from, or denied on the basis of, the abstract state module alone. The periodic exclusion assumes $e_n < a_n$. The bounded-negative state declaration lists strict centring and normalised vanishing as hypotheses; the latter implies the former eventually by taking $K = 1$. Sections 3 and 12 identify the canonical bridge.

B A factorial residue reduction for forced orbits

In the case $m = c = 1$ of Section 9, where the shape equation (5.1) reads $D_n + 1 = (a_n - 1)(n + 1)$, each multiplier is determined by its predecessor; we call such an orbit *forced*. At index n the numerator of the next multiplier is

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the **forced numerator**, and the divisor is $n + 2$. The orbit survives a step when that division is exact, giving a survival predicate, the **survival predicate**. Example 9.1 is the forced orbit from $a = 3$ read this way: $\text{num}(0, 3) = 6$ is divisible by 2 and gives $a_1 = 3$, while $\text{num}(1, 3) = 13$ is not divisible by 3, so the orbit stops there. Deciding survival by iteration is expensive because the orbit grows doubly exponentially, and it is unnecessary: survival over a finite horizon depends on the initial value only through a factorial residue. Computing a pseudo-greedy orbit through residues modulo a shrinking product modulus is Koizumi's method [12, Remark 2 and Algorithm 1, pp. 13–14]; for the forced numerator that product is a factorial.

Theorem B.1 (factorial residue reduction). *For all h and all integers $a \equiv b \pmod{(h+1)!}$, the orbit from a survives h forced updates exactly when the orbit from b does.*

Proof. Let $M(0, i) = 1$ and $M(h+1, i) = (i+2)M(h, i+1)$, an ascending factorial with $M(h, 0) = (h+1)!$. The numerator is a polynomial with integer coefficients, so congruences transfer; reducing the modulus gives divisibility by $i+2$ for one exactly when for the other, and cancelling that common factor from both values and modulus leaves the inductive hypothesis at $M(h, i+1)$. $\square$

Formalised as the **factorial residue reduction**, over the **shrinking-modulus transport**, the **polynomial congruence**, and the **exact-division cancellation**; the modulus identifications are the **ascending-factorial form** and the **factorial value at the initial index**.

At $h = 1$ the modulus is $2! = 2$, and surviving one update means $2 \mid a^2 - 2a + 3$, which holds exactly for odd a : for instance $\text{num}(0, 3) = 6$ but $\text{num}(0, 4) = 11$. So one step of survival is decided by the parity of a alone, which is Theorem B.1 at its smallest nontrivial horizon.

The supplied research record reports a search over initial states below 5000 that produced forced prefixes of length 17 and no longer. That historical computation was not rerun in this review; Theorem 9.2 now excludes the constant-negative case outright, for every seed and at every scale. The reduction is retained because it is exact, and because the shrinking-modulus technique transfers to any forced orbit whose step is a polynomial division.

C Result map and proof dependencies

The following map is for mathematical navigation, not a catalogue of every registered declaration. A checked component does not confer that status on an assembled ordinary theorem.

Bounded defect. Canonical rational tail and normalised vanishing $\rightarrow$ eventual absorption; cofinally bounded negative values $\rightarrow$ stable gcd; reduced tail plus bounded rise $\rightarrow$ CRT contradiction. The state endpoint and the identified canonical R7 transfer have supplied main-repository build evidence. The extra bounded-defect assumption is not deduced from the problem.

Finite negative mass. $C_{n+1} \leq C_n(1 + (-E_n)_+/C_n)$ and summability $\rightarrow$ bounded integer height $\rightarrow$ finitely many rises and eventual constancy. This scalar result needs no normalised vanishing. Its scalar and exact-orbit endpoints are among the cited main-repository declarations; the separately linked R8 canonical formulation is outside that receipt.

Weighted records and protected epochs. LCM records give weighted first crossings; primitive prime-power persistence gives protected epochs. The canonical odd-power supply and the assembled energy equivalence are proved ordinarily in this revision. Separate-release integer lemmas and main-repository conditional record declarations retain their own, narrower evidence boundaries.

Cubic rate. The quantitative tail comparison and Gamma-ratio estimate $\rightarrow$ integer cubic extraction; transport congruences and Chebotarëv $\rightarrow$ a square in the cubic field; trace classification $\rightarrow$ two scales, each obstructed modulo seven. The displayed assembly is an ordinary proof. Finite checks and component sources are supporting evidence, not a formalised cubic-rate endpoint.

Static barriers. Sparse primes disprove a scale-free small-rise extension. At the specified double-exponential modulus scale, prefix density and CRT give the exact whole-modulus maximal-gap coefficient. Both are ordinary auxiliary results, without a canonical reciprocal-tail realisation.

What remains. The unrestricted problem would require an additional global estimate, such as the stated mass or record budget, or a contradiction to the full nonterminal profile. None of the static examples or local identities supplies one. The more detailed source registry and inherited declaration catalogue remain in the research package rather than being repeated here.

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